RAJALAKSHMI ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB MANUAL

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

VISION & MISSION

OF

COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

Vision

To promote highly ethical and innovative computer professionals through excellence in teaching, training and research.

Mission

● To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.

● To promote research activities amongst the students and the members of faculty that could benefit the society. ● To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of networking, security, data mining, web technology, mobile communication and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POs)

PO1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration forthe public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety,

legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and normsof the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with theengineering community and with society at large, such as, being able to comprehend and write effectivereports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one‟s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage inindependent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Computer Science and Engineering Program will demonstrate,

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areasrelated to algorithms, system software, web design, machine learning, data analytics, and networking for efficient design of computer-based systems of varying complexity. Familiarity and practicalcompetence with a broad range of programming language and open source platforms.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computationaltask, model real world problem using appropriate data structure and suitable algorithm. To understand the standard practices and strategies in software project development, using open- ended programmingenvironments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify researchgaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible computer science professional.

Objectives:
• To understand the fundamental concepts of data science.
• To understand data cleaning and processing for data science.
• To learn the fundamentals of exploratory data analysis (EDA).
• To understand inferential data analytics
• To understand the Machine Learning concepts.

Course Outcomes: On completion of course, students will be able to
• Explore the fundamental concepts of data science

• Explain how data is collected, managed and processed for data science
• Understand the fundamentals of exploratory data analysis.
• Perform Statistical Inference and Analyze Variance in Data
• Understand various machine learning algorithms used in data science process.

CO - PO – PSO matrices of course

COs/POs&PSOs	PO1	PO2	PO3	PO4	PO5	PO 6	PO7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO3
CS23334.1	2	2	1	2	2	-	-	-	1	1	1	2	2	2	2
CS23334.2	2	1	-	1	1	-	-	-	2	1	1	2	2	3	1
CS23334.3	2	2	1	2	2	1	1	-	1	2	1	3	2	2	3
CS23334.4	3	2	2	1	2	-	-	-	1	1	2	2	3	3	2
CS23334.5	2	2	1	2	2	-	-	-	1	1	1	2	2	2	2
Average	2.2	1.8	1.25	1.6	1.8	1	1	-	1.2	1.2	1.2	2.2	2.2	2.4	2

3

S.No	List Of Experiments	CO
1.a	Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.	1
1.b	Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.	1
1.c	Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.	1
1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	1
2	Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.	2
3.a	Conduct an experiment to show data visualization using line plot	2
3.b	Conduct an experiment to show data visualization using bar chart	2
3.c	Conduct an experiment to show data visualization using pie chart	2
4	To Count the frequency of occurrence of a word in a body of text is often needed during text processing.	1
5	Data Collection and Initial Exploration	2
6	Time Series Analysis of Website Traffic	5
7	Random Sampling and Sampling Distribution	4
8	Conduct Z test for the Data given using Python	4

9	Experiment to understand Matplotlib library use cases in Data Science through visualization.	1
10	Experiment to understand array function in Data science.	1
11	Experiment to understand pandas library use cases in Data science.	1
12	Experiment to detect outliers in a given data set.	2
13	Experiment to handle missing values and inappropriate data in a given data set.	2
14	Experiment to understand feature scaling	3
15	Experiment to understand the data preprocessing in Data science	2
16	Experiment to understand EDA-Quantitative and Qualitative analysis.	3
17	Experiment to understand Linear Regression for a given data set.	5
18	Experiment to understand KNN algorithm for a given data set.	5
19	Experiment to understand Logistic Regression for a given data set.	5
20	Experiment to understand K-means clustering algorithm for a given data set.	5

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Exp No:**1.a** Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
```

```
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number ofJob Postings')
plt.show()
```

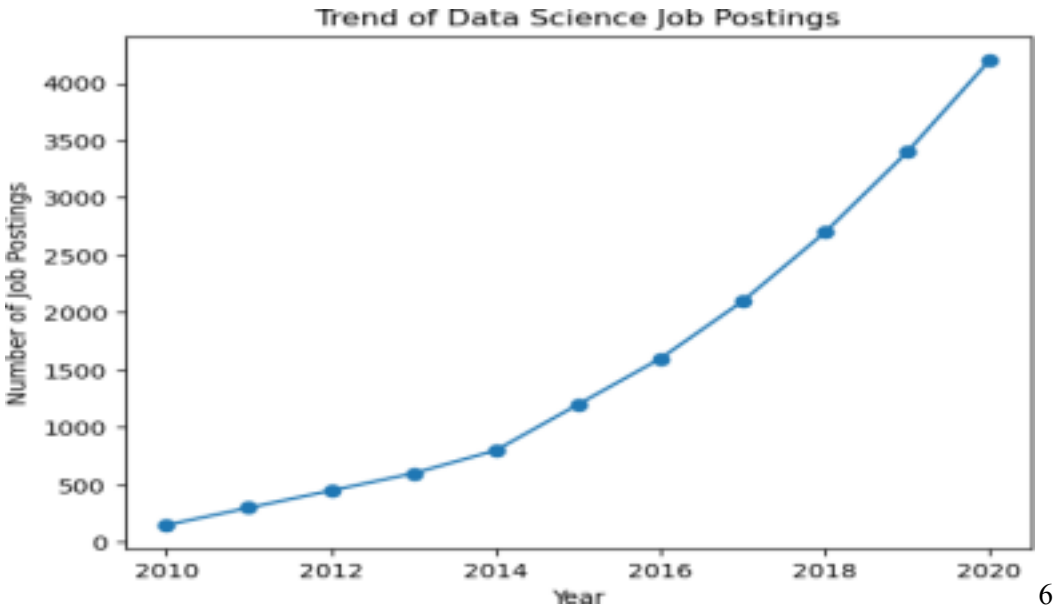
5

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Sample Output:



6

Exp No:**1.b** Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer,

Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

plt.show()
```

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst. counts = 300, 500, 450, 200, 150.

Sample Output:



7

Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

```
# Structured data example

structured_data = pd.DataFrame({
```

```
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,  or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}  print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

```
Structured Data:
ID Name Age
0 1 Alice 25
1 2 Bob 30
2 3 Charlie 35

Unstructured Data:
This is an example of unstructured data. It can be a piece of text, an  image, or a video file.

Semi-structured Data:
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68} }
```

8

Exp No:**1.d** Conduct an experiment to encrypt and decrypt given sensitive data. **Description:** Use the cryptography library to encrypt and decrypt a piece of data.

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
```

```
f.decrypt(token)

b'Rajalakshmi Engineering College'

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```


Exp No:**2** Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame

file_path='C:\sales_data.csv'

df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
```

```
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
print(product_summary)
```



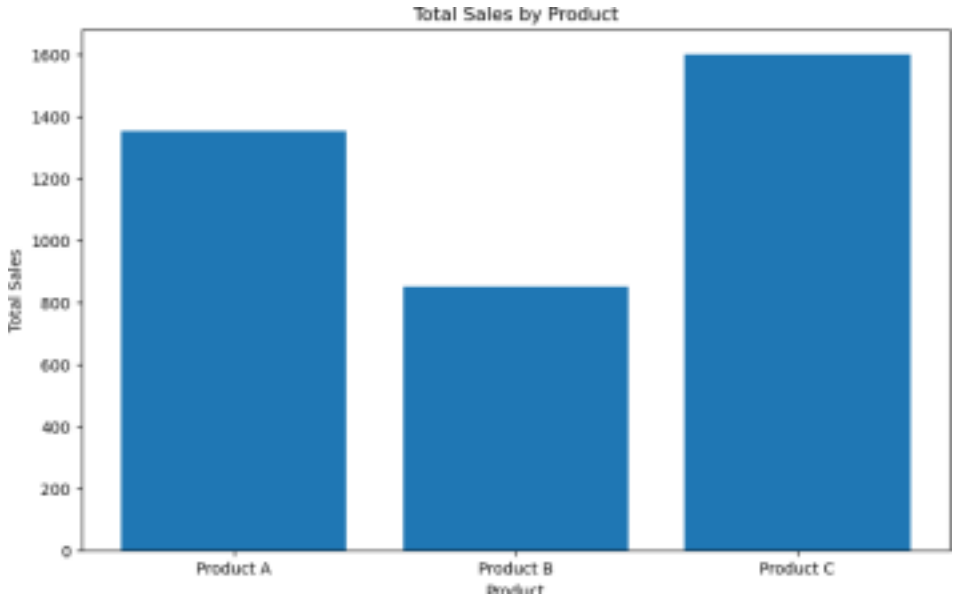
```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])  plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()  plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
```

```
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',  aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')  plt.title('Correlation Matrix')
plt.show()
```

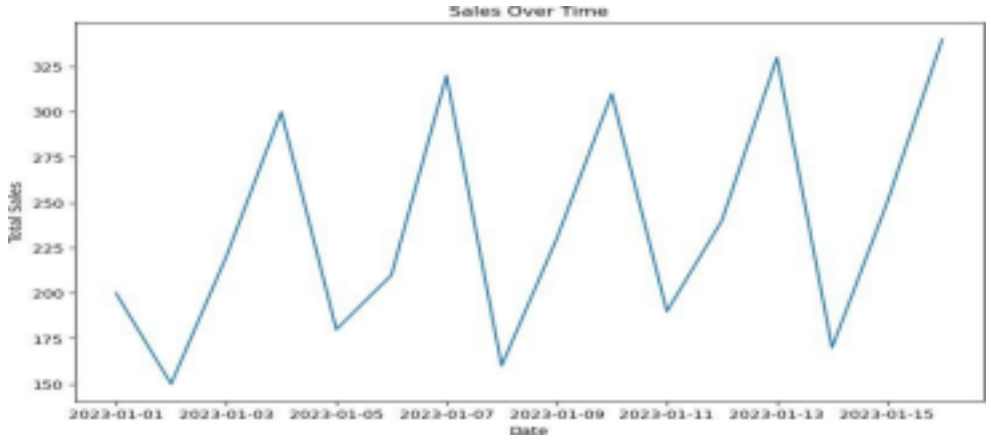
11

OUTPUT:

```
Date Product Sales Quantity Region
0 2023-01-01 Product A 200 4 North
1 2023-01-02 Product B 150 3 South
2 2023-01-03 Product A 220 5 North
3 2023-01-04 Product C 300 6 East
4 2023-01-05 Product B 180 4 West
Date 0
Product 0
Sales 0
Quantity 0
Region 0
dtype: int64
Sales Quantity
count 16.000000 16.000000
mean 237.500000 5.375000
std 64.031242 1.746425
min 150.000000 3.000000
25% 187.500000 4.000000
50% 225.000000 5.500000
75% 302.500000 7.000000
max 340.000000 8.000000
Product Sales Quantity
0 Product A 1350 33
1 Product B 850 17
2 Product C 1600 36
```



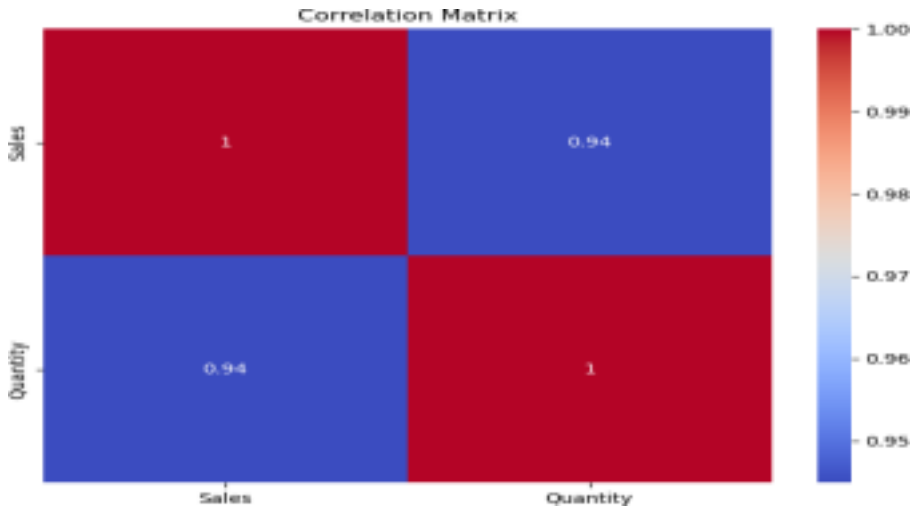
12




```
Product Product A Product B Product C
Region
East 0 0 1600
North 1350 0 0
South 0 480 0
West Sales

0
Sales
1.000000
370
Quantity  0.944922
0


Quantity 0.944922 1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur  eWarning: The default value of numeric_only in DataFrame.corr is deprecated . In a future version, it will default to False. Select only valid columns  or specify the value of numeric_only to silence this warning.
correlation_matrix = df.corr()
```



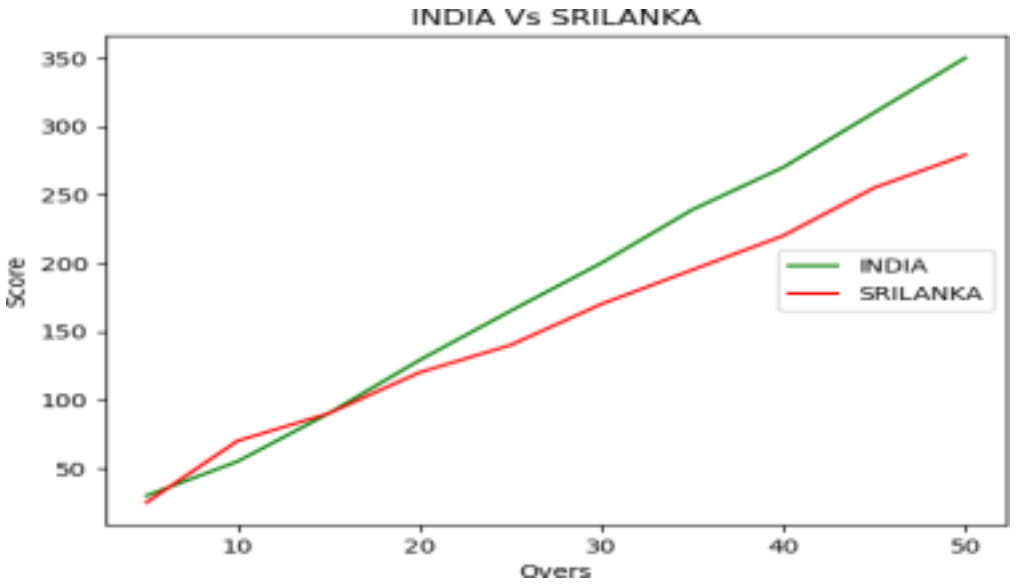
13

Exp No:**3.a** Conduct an experiment to show data visualization using line plot .

Description: Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as cricket
Overs=list(range(5,51,5))
Indian_Score=[30,55,90,129,165,200,239,270,310,350]
Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]
cricket.plot(Overs,Indian_Score)
cricket.plot(Overs,Srilankan_Score)
cricket.show()
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend()
cricket.plot(Overs,Indian_Score,color="green",label="INDIA")
cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")
cricket.legend(loc="center right")
```

OUTPUT:

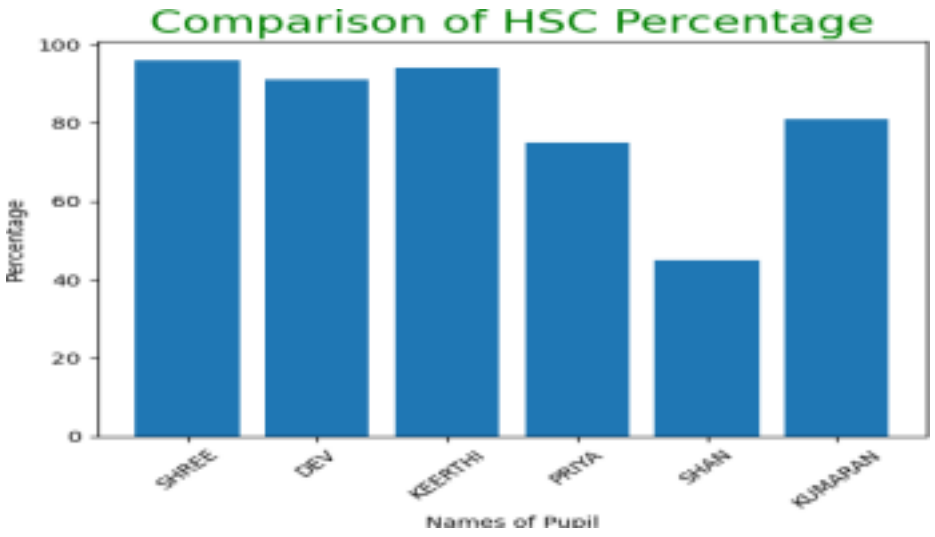


Exp No:**3.b** Conduct an experiment to show data visualization using bar chart .

Description: Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

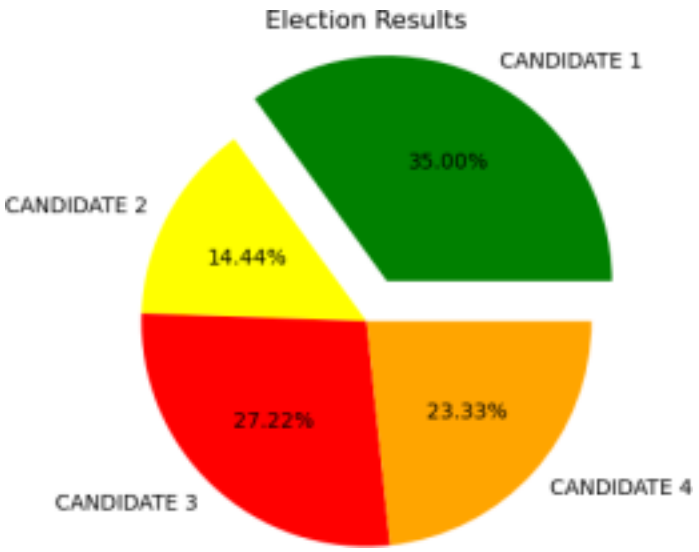
OUTPUT:



Exp No:3.c Conduct an experiment to show data visualization using pie chart .**Description:** Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as election
# Election data
labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']
Votes = [315, 130, 245, 210]
colors = ['green', 'yellow', 'red', 'orange']
explode = (0.2, 0, 0, 0)
# Plotting the pie chart
election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')  election.title('Election Results')
election.show()
```

OUTPUT:



Exp No:**4** To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

Description: Import the word_tokenize function and gutenberg.

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt') # Ensure the punkt tokenizer models are also downloaded  # Load the text from the gutenberg
corpus
sample = gutenberg.raw("austen-emma.txt") # Change to an existing text from the corpus  # Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

```
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1),
('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma ', 2), ('Woodhouse', 1),
(',', 5), ('handsome', 1), (',', 5), ('clever' , 1), (',', 5), ('and', 3), ('rich',
1), (',', 5), ('with', 2), ('a', 1 ), ('comfortable', 1), ('home', 1), ('and', 3),
('happy', 1), ('disposi tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1),
('some', 1 ), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e
xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly ', 1),
('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world' , 1), ('with', 2)]
```


Exp No:5 Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset. **Materials Needed:**

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. **Data Collection:**

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

```
[1] import pandas as pd
    db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

2. Initial Exploration:


```
# Dataset structure and summary statistics
print(db.info())
print(db.describe())

# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

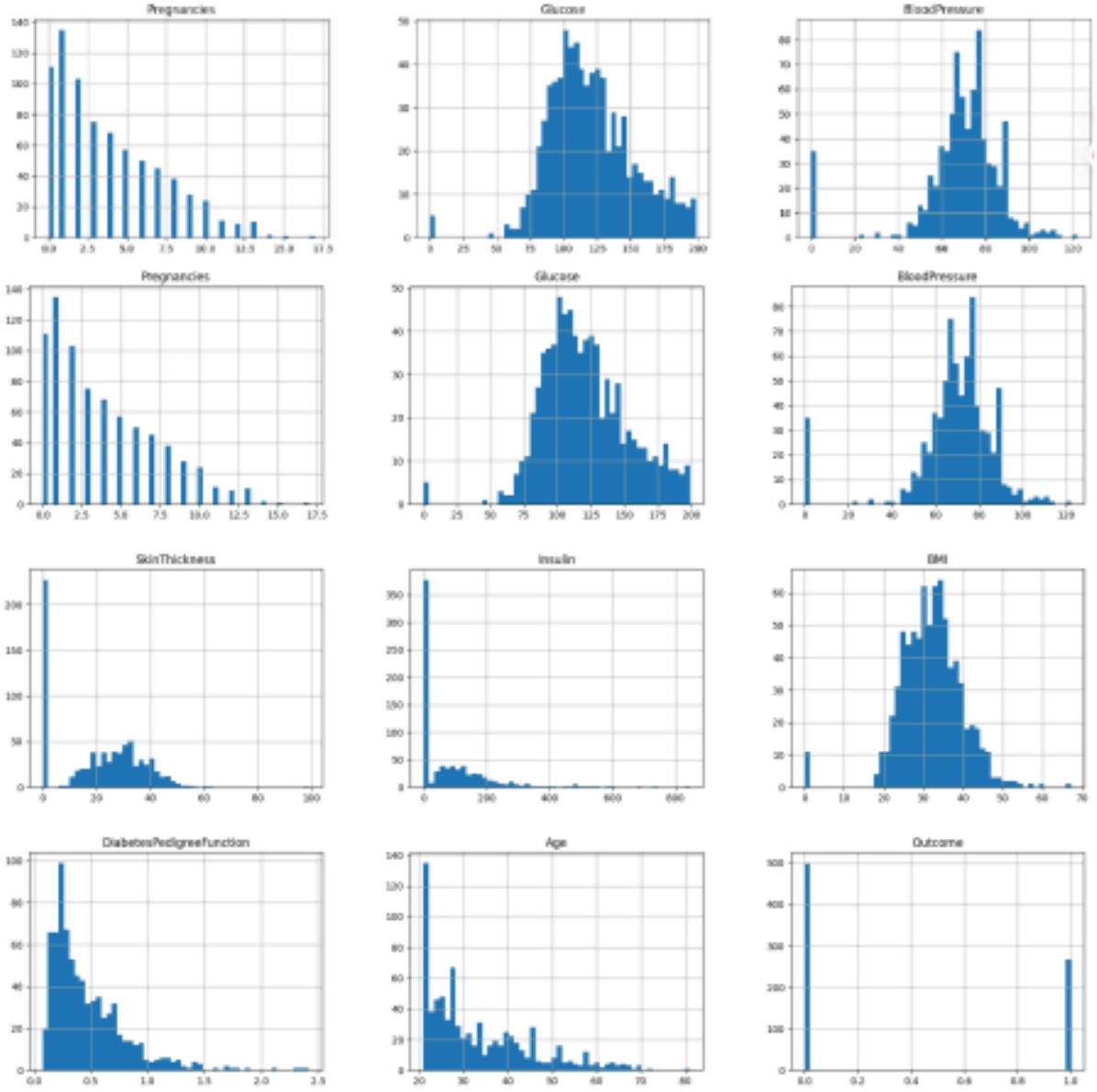
db.hist(bins=50, figsize=(20, 15))
plt.show()

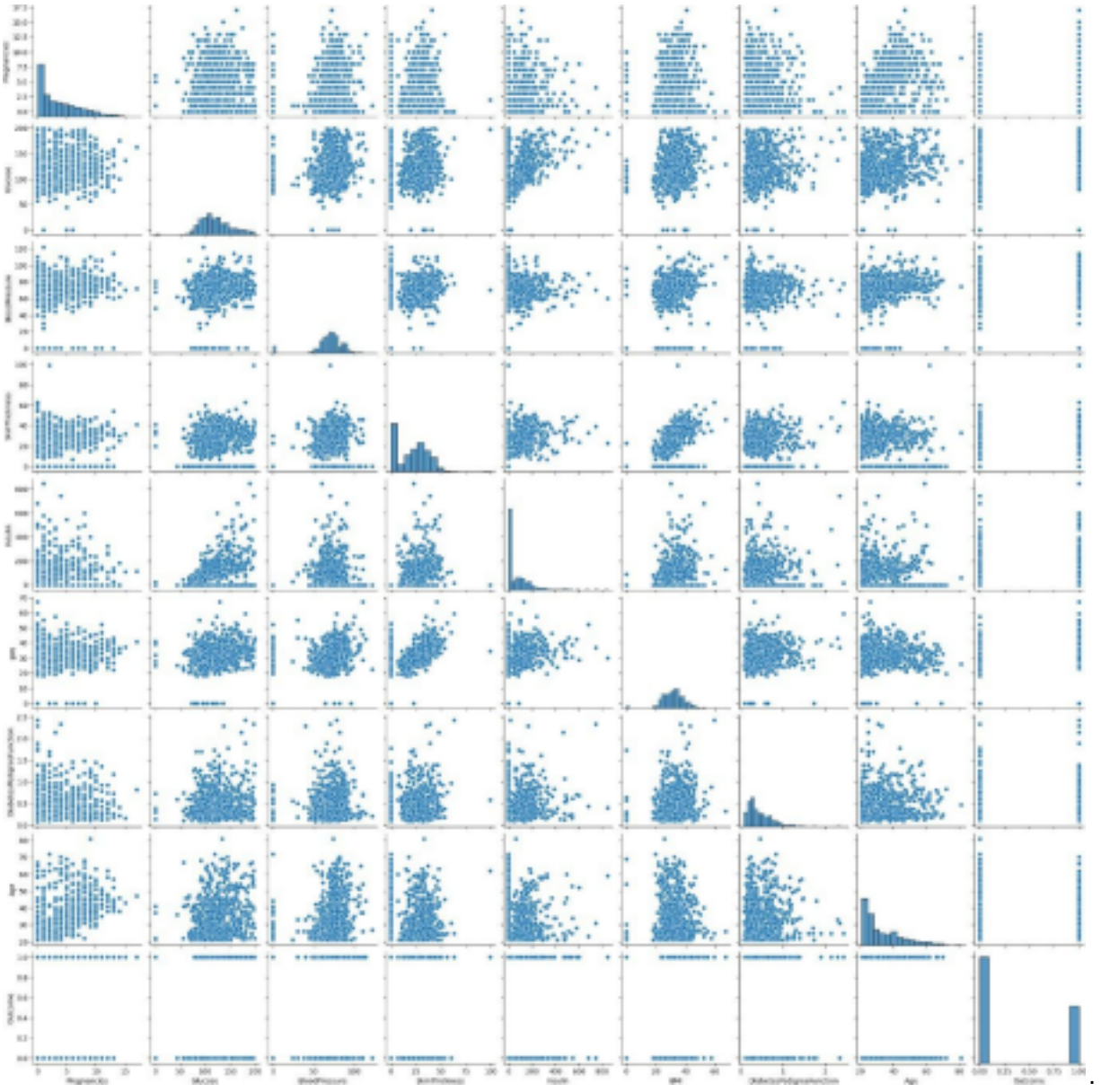
# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()
```

[3]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000   
mean      3.845052  120.894531      69.105469      20.536458   79.799479   
std       3.369578   31.972618      19.355807      15.952218  115.244002   
min       0.000000    0.000000       0.000000       0.000000    0.000000   
25%       1.000000   99.000000      62.000000       0.000000    0.000000   
50%       3.000000  117.000000      72.000000      23.000000   30.500000   
75%       6.000000  140.250000      80.000000      32.000000  127.250000   
max      17.000000  199.000000     122.000000      99.000000  846.000000   
```

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000





21

Experiment 6: Time Series Analysis of Website Traffic

Objective: To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset: A dataset containing daily website traffic data with columns such as `Date`, `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.

Methodology:

1. **Load and Clean Data:**
   - Import the dataset and parse the `Date` column as a datetime object.
   - Handle missing values by interpolation.

2. **Time Series Decomposition:**
o Decompose the time series data of `Page_Views` to identify trend, seasonality, and residuals.
3. **Visualization:**
o Plot the time series data for `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.
o Use moving averages to smooth the time series and highlight trends. o Create seasonal plots to visualize patterns over time.
4. **Anomaly Detection:**
o Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

Sample code

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose



# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')



# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')  decomposition.plot()
```



```
plt.show()



# Moving Average

data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()



# Anomaly Detection

data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()
```

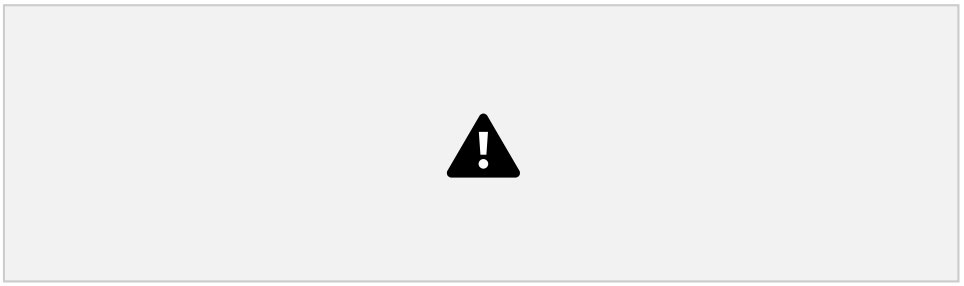
```
anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')  plt.legend()

plt.show()
```

sample output



Expected Outcomes:

· Clear visualization of trends, seasonality, and anomalies in website traffic. · Understanding of the underlying patterns in user behavior over time.

Experiment 7:. Random Sampling and Sampling Distribution

Objective:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. **Generate a Population:**
○ Create a population of data with a specified distribution (e.g., normal distribution). 2. **Random Sampling:**

○ Perform random sampling from the population to create multiple samples of different sizes.
○ Compute sample statistics (mean, standard deviation, etc.) for each sample. 3. **Sampling Distribution:**
○ Plot histograms or density plots of sample statistics (e.g., sample means). ○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. **Central Limit Theorem (Optional):**
○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Example Code:

python

Copy code

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)

# Step 2: Random sampling
sample_sizes = [30, 50, 100] # different sample sizes to consider
num_samples = 1000 # number of samples for each sample size
```



```
sample_means = {}

for size in sample_sizes:
sample_means[size] = []
for _ in range(num_samples):
sample = np.random.choice(population, size=size, replace=False)
sample_means[size].append(np.mean(sample))

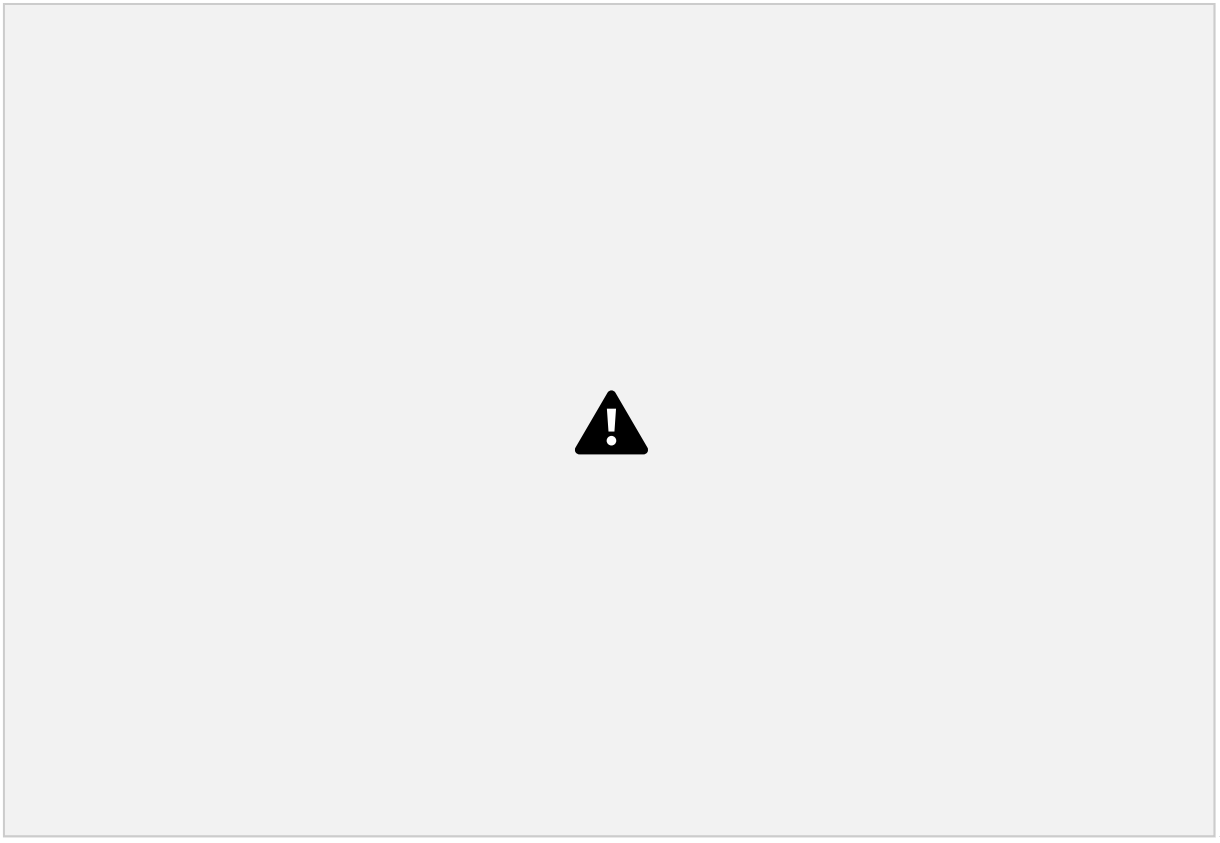
# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
plt.subplot(len(sample_sizes), 1, i+1)
plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
```

```
plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population  Mean')
plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.legend()

plt.tight_layout()
plt.show()
```


OUTPUT:



Lab Experiment 8: Conduct Z test for the Data Given using Python Objective: To test whether the average weight of a species of birds differs from 150 grams. **Procedure:**

1. **Null Hypothesis (H_0):** The average weight of the birds is 150 grams. 2. **Alternative Hypothesis (H_1):** The average weight of the birds is not 150 grams. 3. **Sample:** Measure the weights of 30 birds randomly selected from the population. 4. **Z-Test:** Conduct a Z-test to compare the sample mean to 150 grams. 5. **Decision Rule:** Use a significance level of $\alpha = 0.05$.

Python Code Example:

python

Copy code

```
import numpy as np

import scipy.stats as stats



# Define the sample data (hypothetical weights in grams)

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152, 151, 148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis

population_mean = 150

# Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation



# Number of observations

n = len(sample_data)

# Calculate the Z-statistic

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))



# Calculate the p-value
```



```
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test



# Print results

print(f"Sample Mean: {sample_mean:.2f}")
```

```
print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:

print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")  else:

print("Fail to reject the null hypothesis: There is no significant difference in average weight from  150 grams.")
```

OUTPUT:

```
Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Exp No:**9** Experiment to understand Matplotlib library use cases in Data Science through visualization

Description: Use Iris data set to understand the Matplot lib library

10/14/24, 12:23 PM irispetalsepal.ipynb - Colab

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline


data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Id 150 non-null int64
 1 SepalLengthCm 150 non-null float64
 2 SepalWidthCm 150 non-null float64
```

```
 3 PetalLengthCm 150 non-null float64
 4 PetalWidthCm 150 non-null float64
 5 variety 150 non-null object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

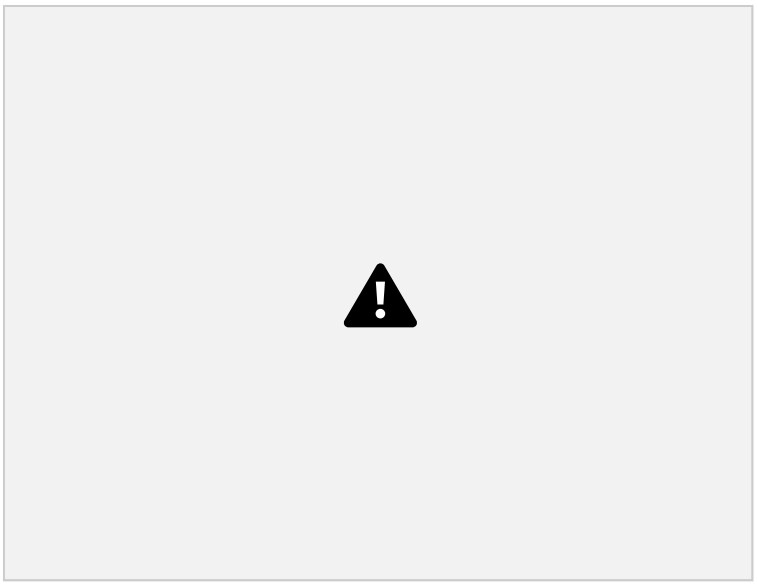
```
data.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150 000000	7 900000	4 400000	6 900000	2 500000

```
data.value_counts('variety')
```

variety	count
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50


```
sns.countplot(x='variety',data=data,)
plt.show()
```



```
dummies=pd.get_dummies(data.variety)
```

```
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
```

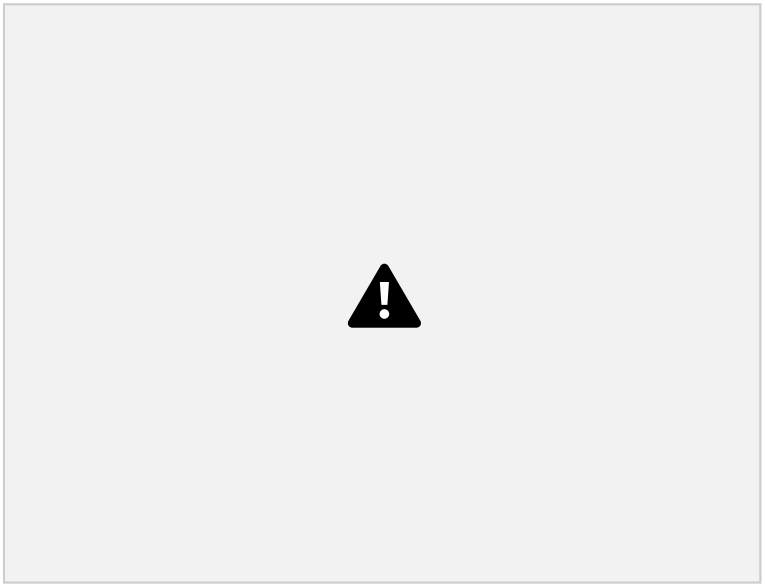
```
FinalDataset.head()
```

	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5	0	3

	6	1	4

```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
```

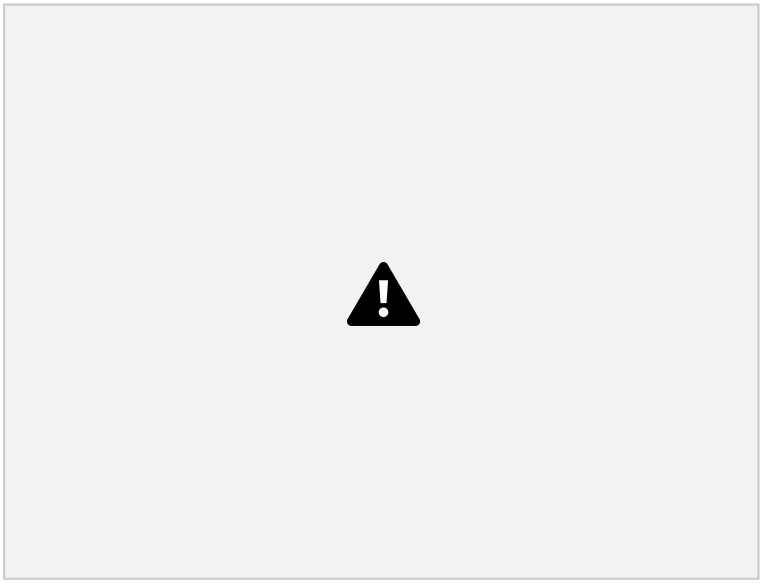
```
<Axes: xlabel='SepalLengthCm', ylabel='SepalWidthCm'>
```



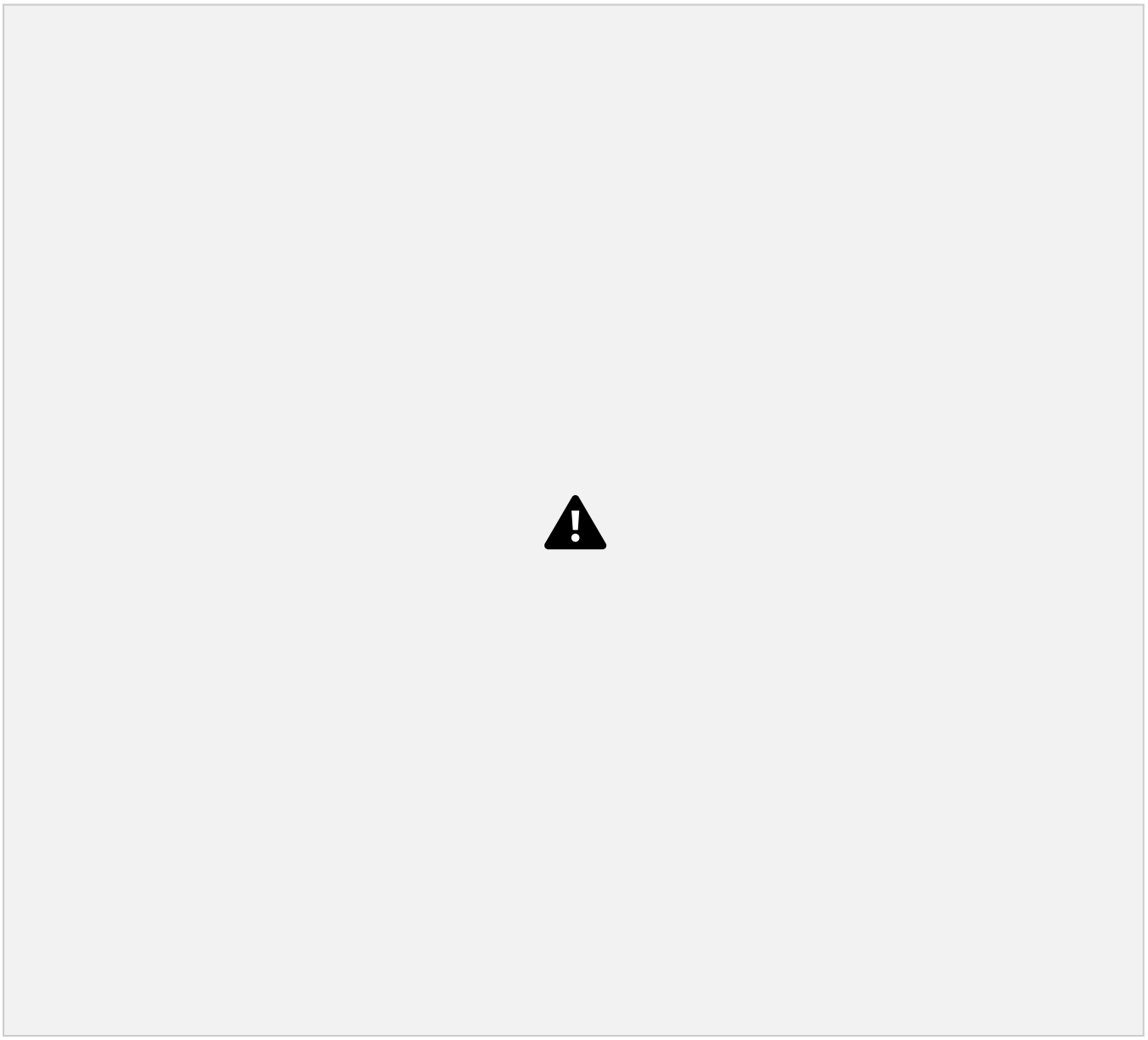
```
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
```



```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```

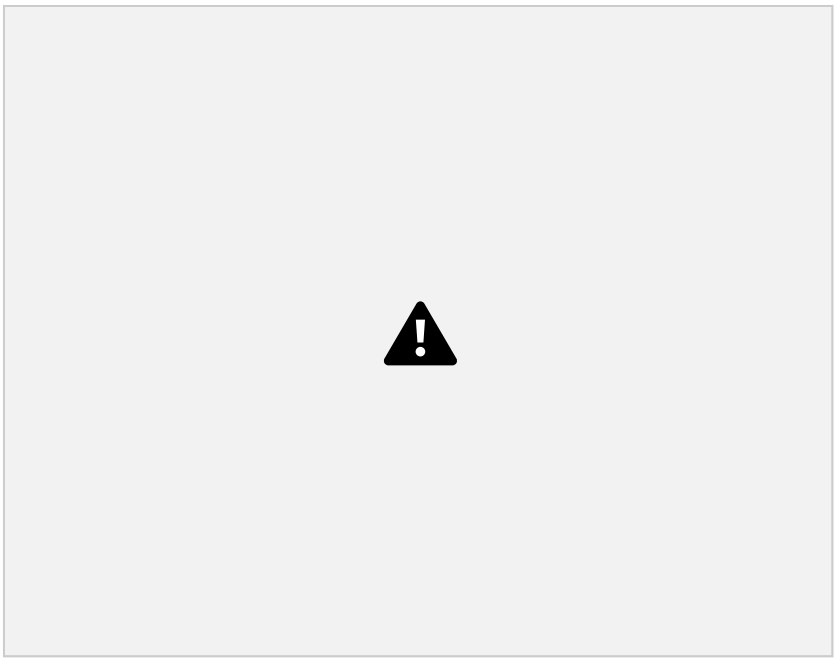


```
sns.pairplot(data,hue='variety',height=3);
```

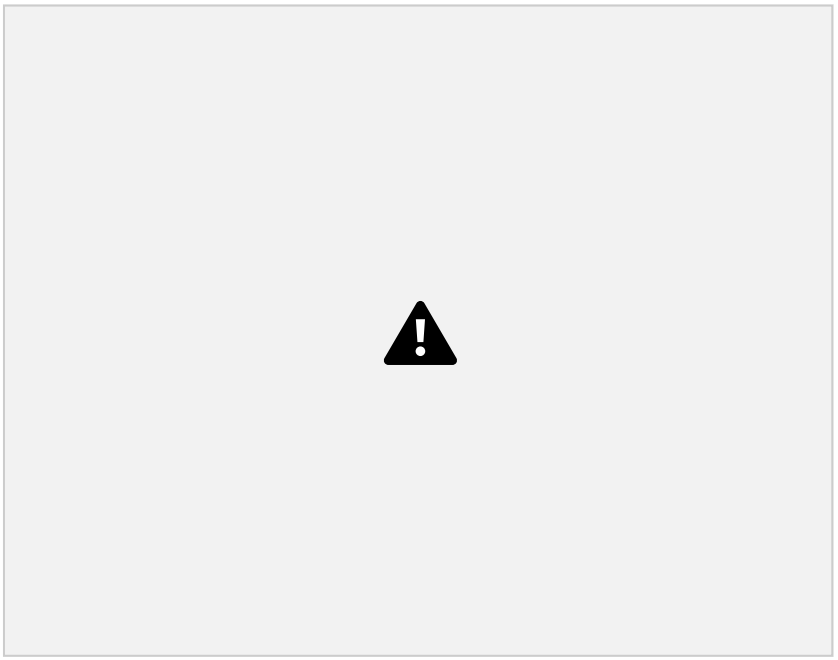


 `plt.show()`

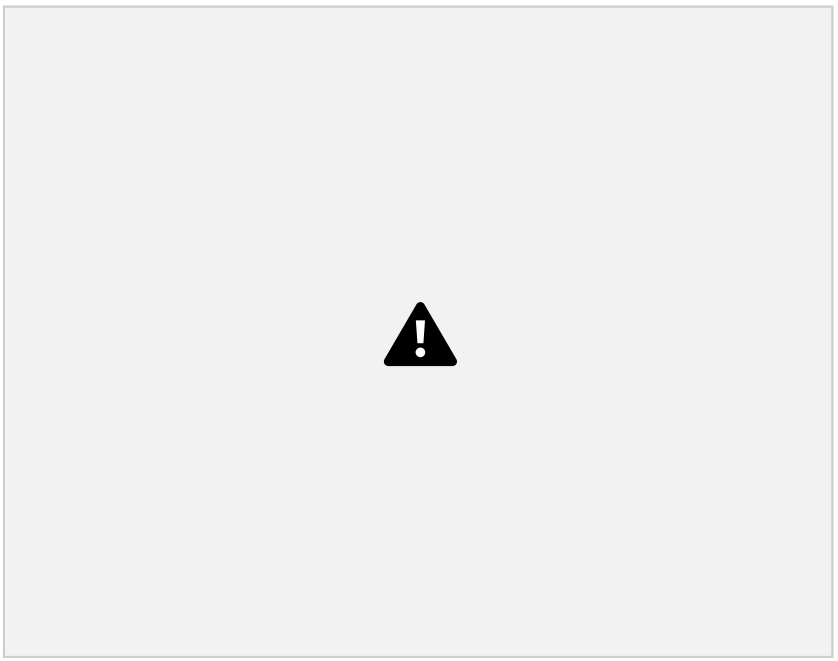
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
```



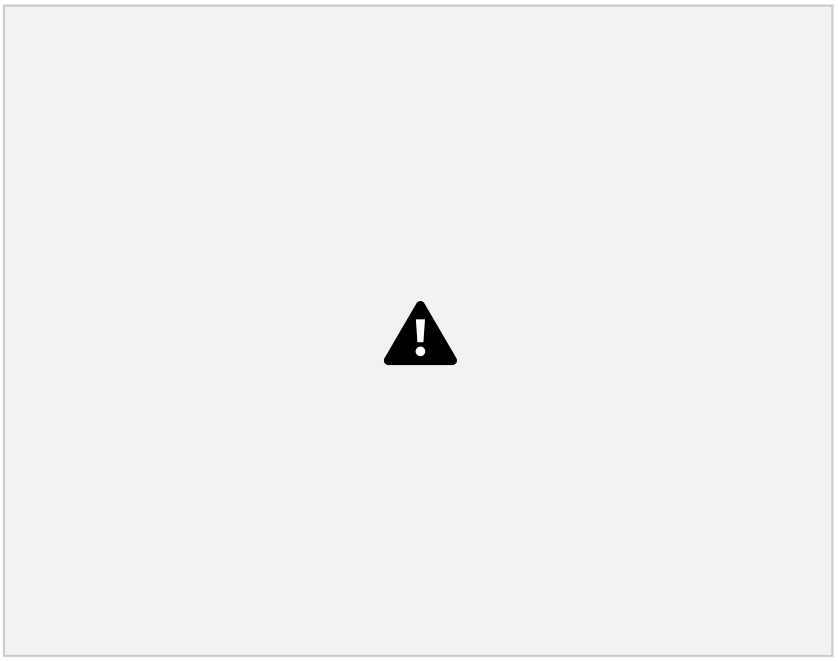
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
```



```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
```

```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();
plt.show();
```



Exp No:**10** Experiment to understand array function in Data science. Description:

Understand array function using Numpy library


```
import numpy as np
array=np.random.randint(1,100,9)
array
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
np.sqrt(array)
```

```
array([9.11043358, 5. , 4.35889894, 6.8556546 , 7.87400787,
       3.87298335, 9.79795897, 6.244998 , 7.14142843])
```

```
array.ndim
```

```
1
```

```
new_array=array.reshape(3,3)
```

```
new_array
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
newm=new_array.reshape(3,3)
```

```
newm
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
newm[2,1:3]
```

```
array([39, 51])
```

```
newm[1:2,1:3]
```

```
array([[62, 15]])
```

```
new_array[0:3,0:0]
```

```
array([], shape=(3, 0), dtype=int64)
```

```
new_array[0:2,0:1]
```

```
array([[83],
       [47]])
```

```
new_array[0:3,0:1]
```

```
array([[83],
       [47],
       [96]])
```

```
new_array[1:3]
```

```
array([[47, 62, 15],
       [96, 39, 51]])
```

Exp No:**11** Experiment to understand pandas library use cases in Data science.

Description: Understand data frame use cases using pandas library


```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
```

```
df=pd.DataFrame(list)
df
```

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

```
df.columns=['Empd','Name','Salary']
df
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Empd 2 non-null int64
 1 Name 2 non-null object
 2 Salary 2 non-null int64
dtypes: int64(2), object(1)
memory usage: 176.0+ bytes
```

```
df=pd.read_csv("/content/50_Startups.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 R&D Spend 50 non-null float64
 1 Administration 50 non-null float64
 2 Marketing Spend 50 non-null float64
 3 State 50 non-null object
 4 Profit 50 non-null float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

```
df.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107 34	91391 77	366168 42	Florida	166187 94

```
df.tail()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0 00	116983 80	45173 06	California	14681 40

10/14/24, 12:15 PM pandasclass.ipynb - Colab

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
```

```
df.head()
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

```
df.tail()
```

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 emp id 7 non-null int64
 1 name 7 non-null object
 2 salary 7 non-null int64
dtypes: int64(2), object(1)
memory usage: 296.0+ bytes
```

```
df.salary
```

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

```
type(df.salary)
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
  th d f d h b idd t t ti ll l d
```

```
df.salary.mean()
```

```
5714.285714285715
```



```
df.salary.median()
```

```
6000.0
```

```
df.salary.mode()
```

	salary

0	5000
1	6000

```
df.salary.var()
```

```
2571428.5714285714
```

```
df.salary.std()
```

```
1603.5674514745463
```

```
df.describe()
```

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7 000000	8000 000000

```
df.describe(include='all')
```

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000
max	7 000000	NaN	8000 000000

```
empCol=df.columns
```

```
empCol
```

```
Index(['emp id', 'name ', 'salary'], dtype='object')
```

```
emparray=df.values
```

```
emparray
```

```
array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
       [3, 'SREYA G', 7000],
       [4, 'SREYASKARI MULLAPUDI', 5000],
       [5, 'SRI AKASH U G', 8000],
```



```
       [6, 'SRI HARSHAVARDHANAN R', 3000],
       [7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
```

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
```

```
employee_DF
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000

4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
Start coding or generate with AI.
```


Exp No:**12** Experiment to detect outliers in a given data set. Description:

Understand the procedure to identify the outliers in a given dataset


```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 to 100
array
```

```
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
```

```
array.mean()
```

```
50.5
```

```
np.percentile(array,25)
```

```
26.0
```

```
np.percentile(array,50)
```

```
56.0
```

```
np.percentile(array,75)
```

```
69.0
```

```
np.percentile(array,100)
```

```
90.0
```

```
#outliers detection
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur
```

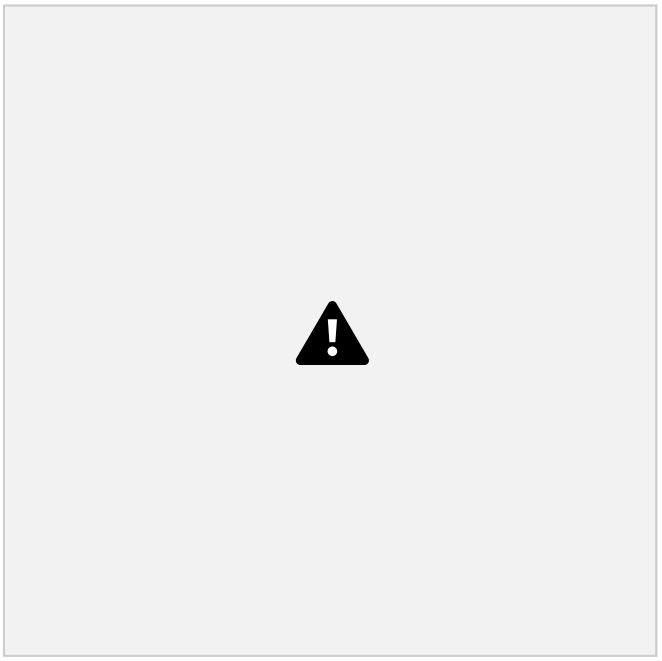
```
lr,ur=outDetection(array)
```

```
lr,ur
```

```
(-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```



```
sns.distplot(array)
```



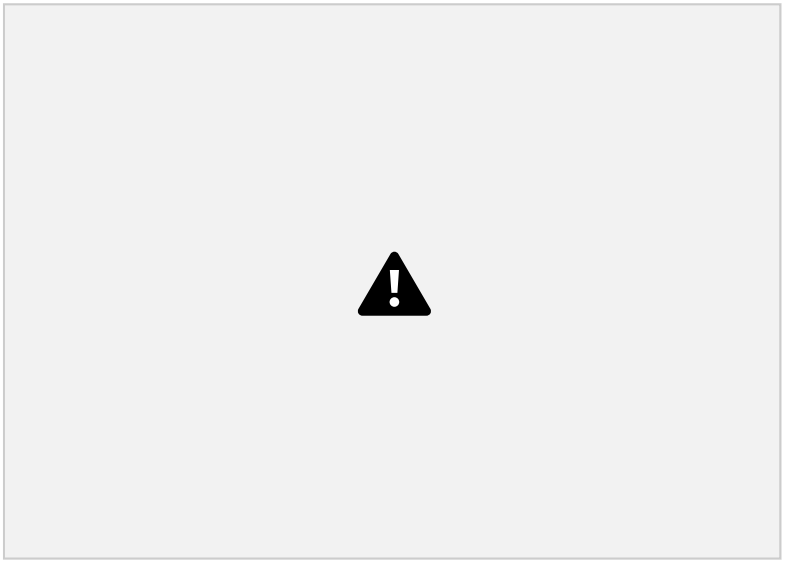
```
<ipython-input-19-d72101983c40>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

 sns.distplot(array)
<Axes: ylabel='Density'>
```

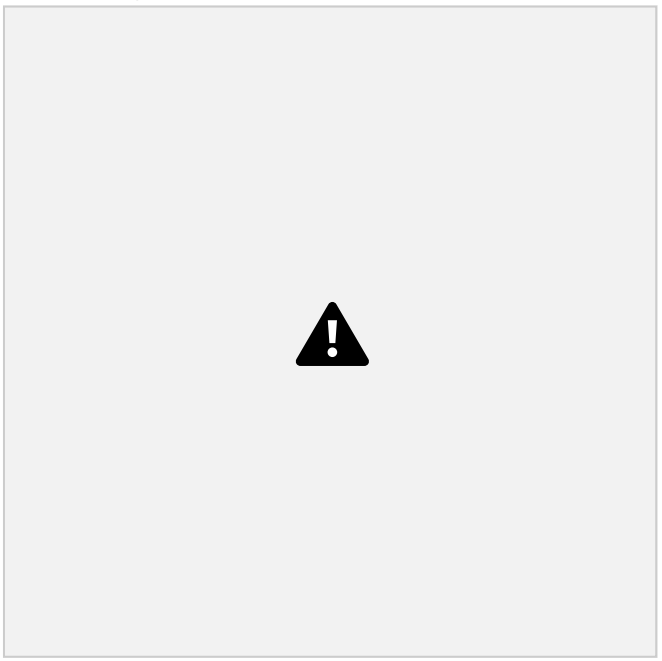


```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
```

```
sns.displot(new_array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>
```



```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

```
(-38.5, 133.5)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

```
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
```



```
sns.distplot(final_array)
```

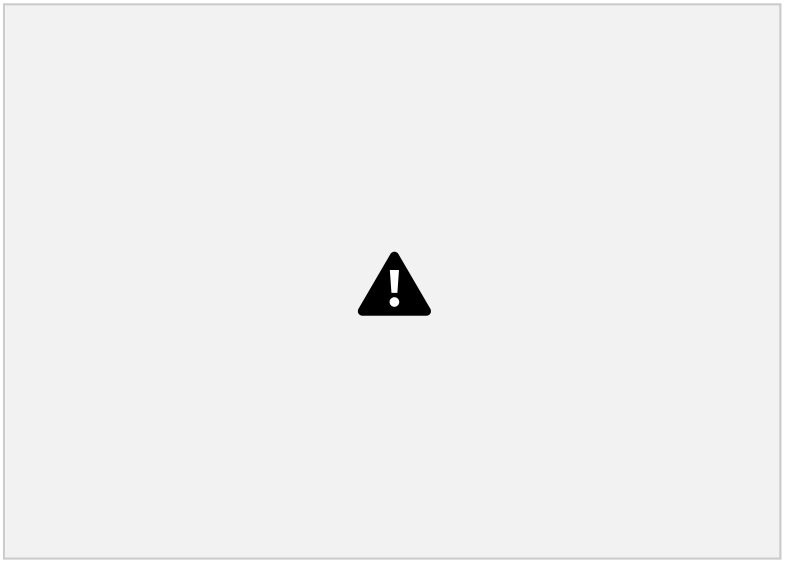
```
<ipython-input-18-7ba96ada5b76>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

 sns.distplot(final_array)
<Axes: ylabel='Density'>
```



Exp No:**13** Experiment to handle missing values and inappropriate data in a given data set

.Description: Understand the procedure to handle missing and inappropriate data

Exp:6 Handling Missing and Inappropriate Data in a Dataset

Aim: Demonstrate an experiment to handle missing data and inappropriate data in a Data set using Python Pandas Library for Data Preprocessing.

Dataset Given:

Hotel.csv

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group
1	20-25	4	Ibis	veg	1300	2	40000	20-25
2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
3	25-30	6	RedFox	Veg	1322	2	30000	25-30
4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
5	35+	3	Ibis	Vegetarian	989	2	45000	35+
6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

About Dataset:

No.of Columns =9 (called as series – CustomerID, Age_Group, Rating(1-5),Hotel, FoodPreference, Bill, NoOfPax, EstimatedSalary)

CutomerID: Numerical Continuous data

Age: Categorical Data

Rating (1-5): Numerical Discrete Data

Hotel: Categorical Data

Food: Categorical Data

Bill: Numerical Continuous data

NoOfPax: Numerical Discrete

EstimatedSalary: Numerical Continuous data

Python Code:

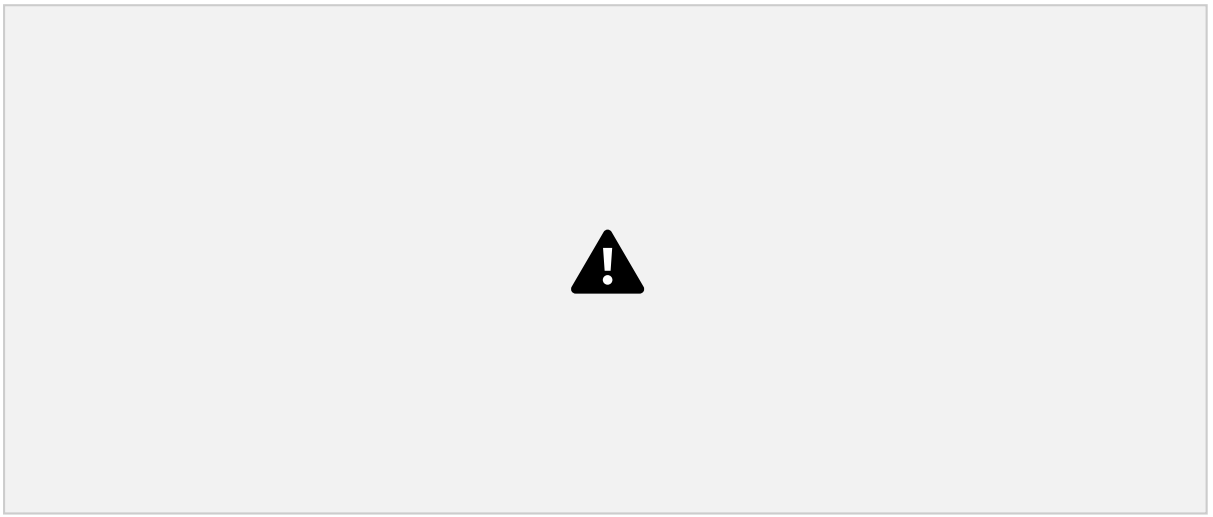
Upload Hotel.csv and convert it into dataFrame

```
import numpy as np

import pandas as pd

df=pd.read_csv("Hotel_Dataset.csv")

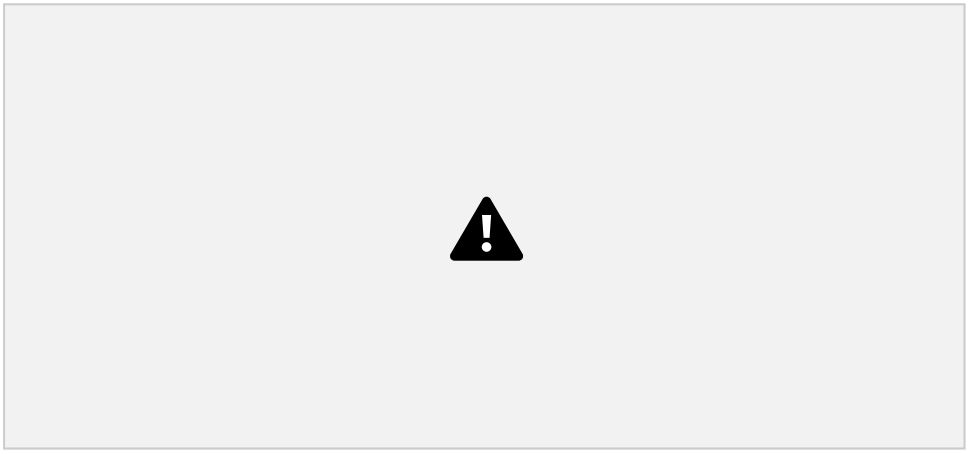
df
```



#From the dataframe identify the duplicate row(i.e row 9)

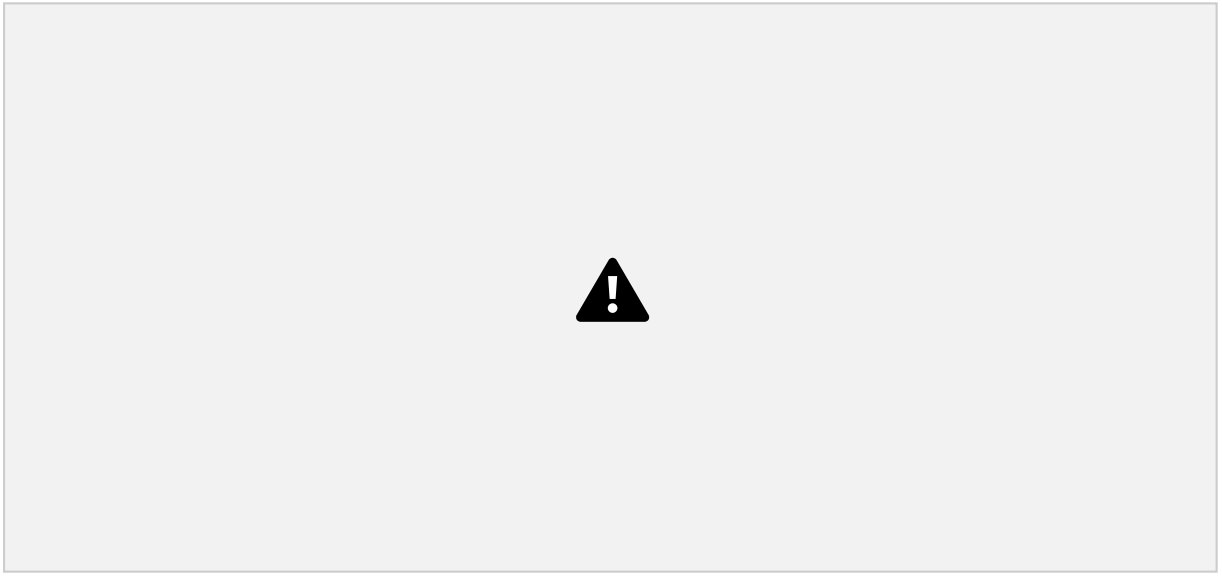
The duplicated() method returns a Series with True and False values that describe which rows in the DataFrame are duplicated and not.

df.duplicated()



The info() method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values).

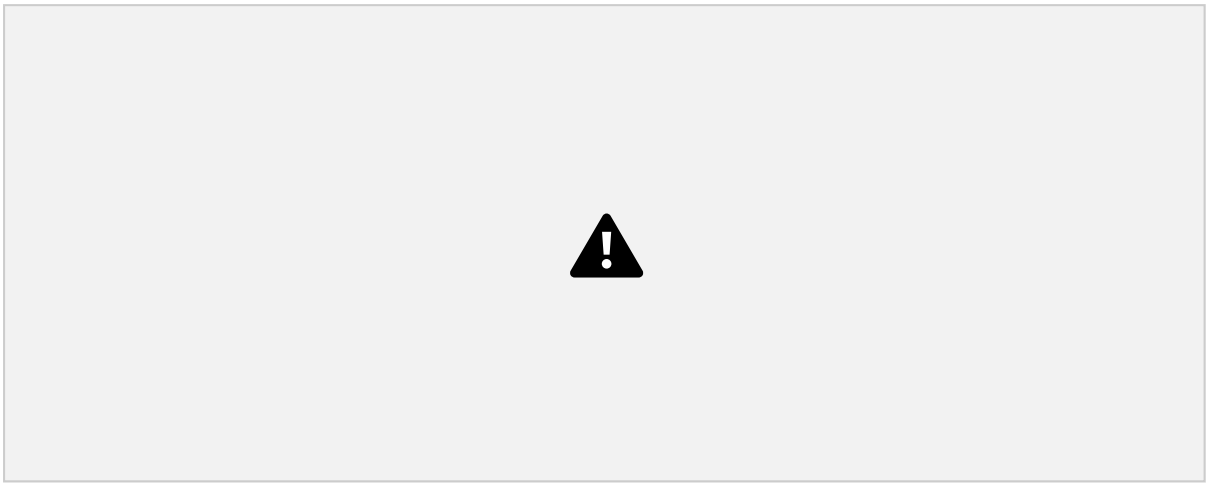
df.info()



The drop_duplicates() method removes duplicate rows.

df.drop_duplicates(inplace=True)

df



#While removing duplicate record row index also removed

The len() function to return the length of an object. With a dataframe, the function returns the number of rows.

len(df)

10

#Reset the index

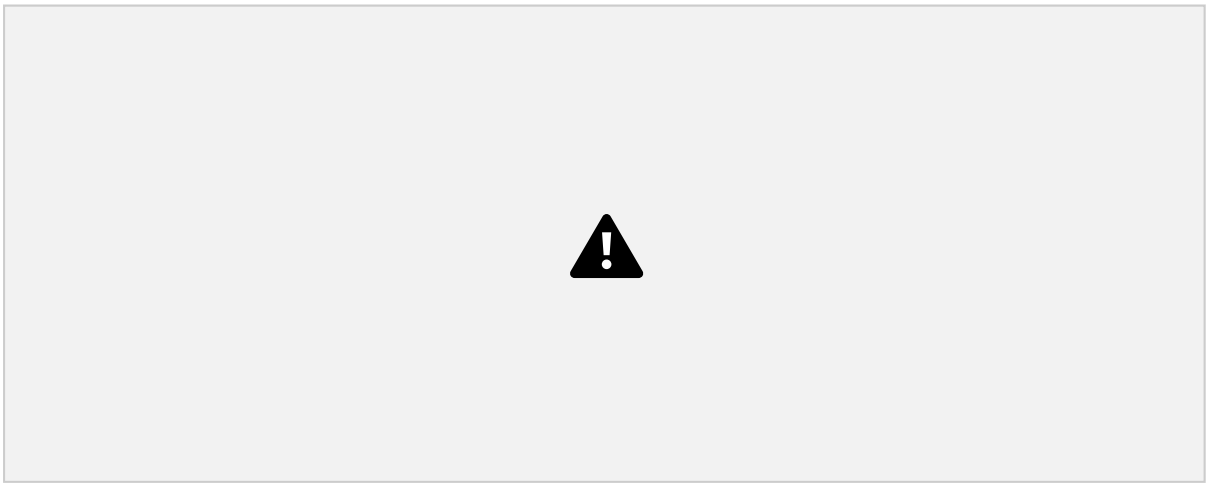
index=np.array(list(range(0,len(df))))

df.set_index(index,inplace=True)

index

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

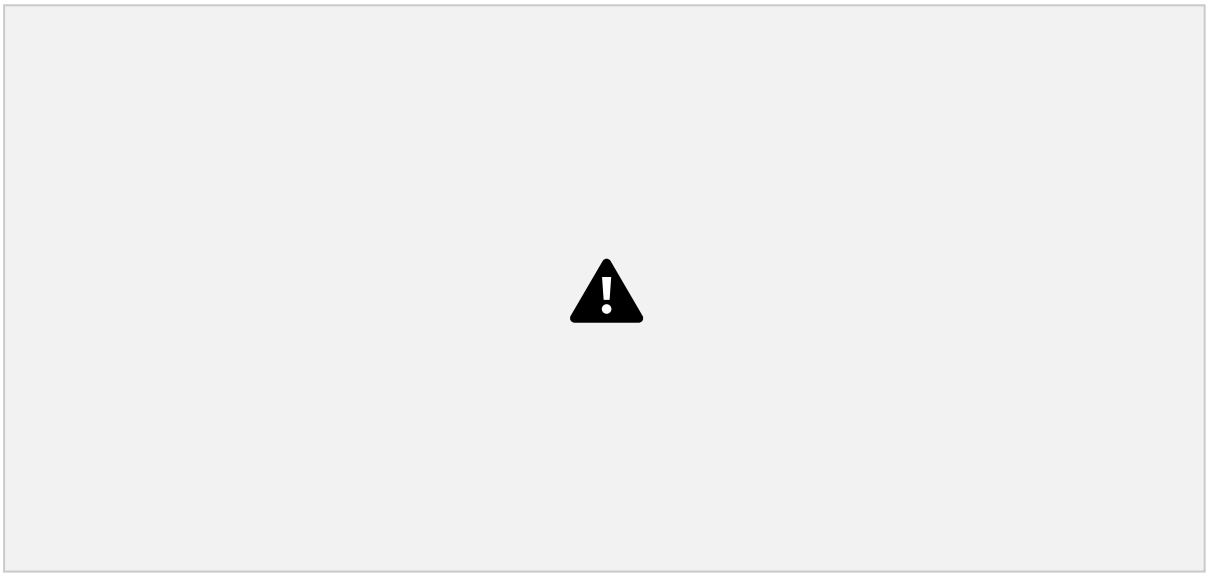
df



Axis refers to the dimensions of a DataFrame (index and columns) or Series (index only) Use axis=0 to apply functions row-wise along the index. Use axis=1 to apply functions column-wise across columns.

df.drop(['Age_Group.1'],axis=1,inplace=True)

df



The function . loc is typically used for label indexing and can access multiple columns.

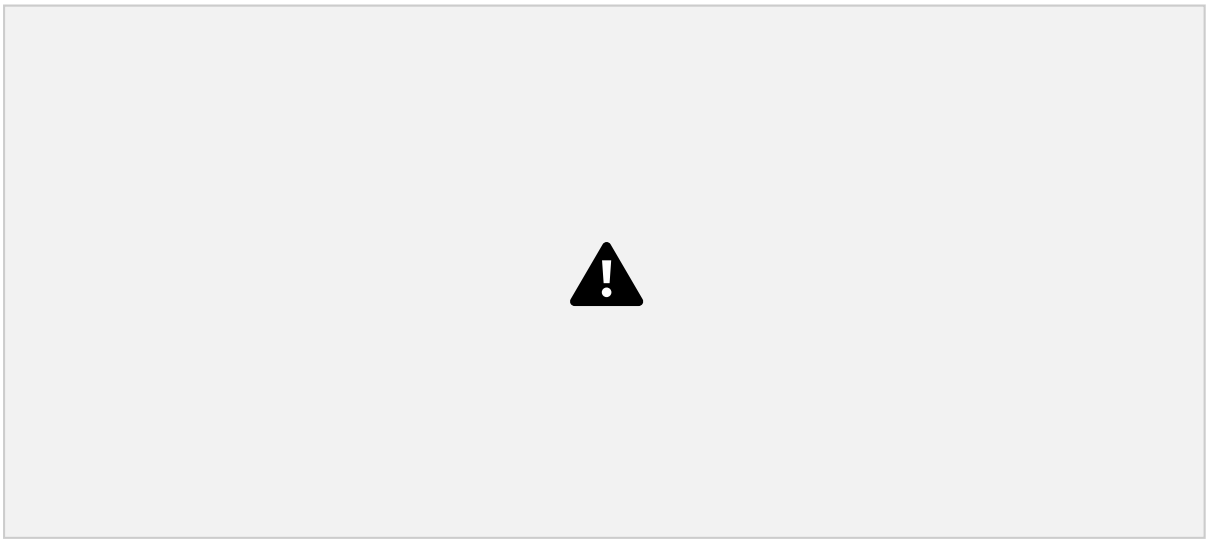
df.CustomerID.loc[df.CustomerID<0]=np.nan

df.Bill.loc[df.Bill<0]=np.nan

df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame See
the caveats in the documentation: https://pandas.pydata.org/pandas
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:2: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
```



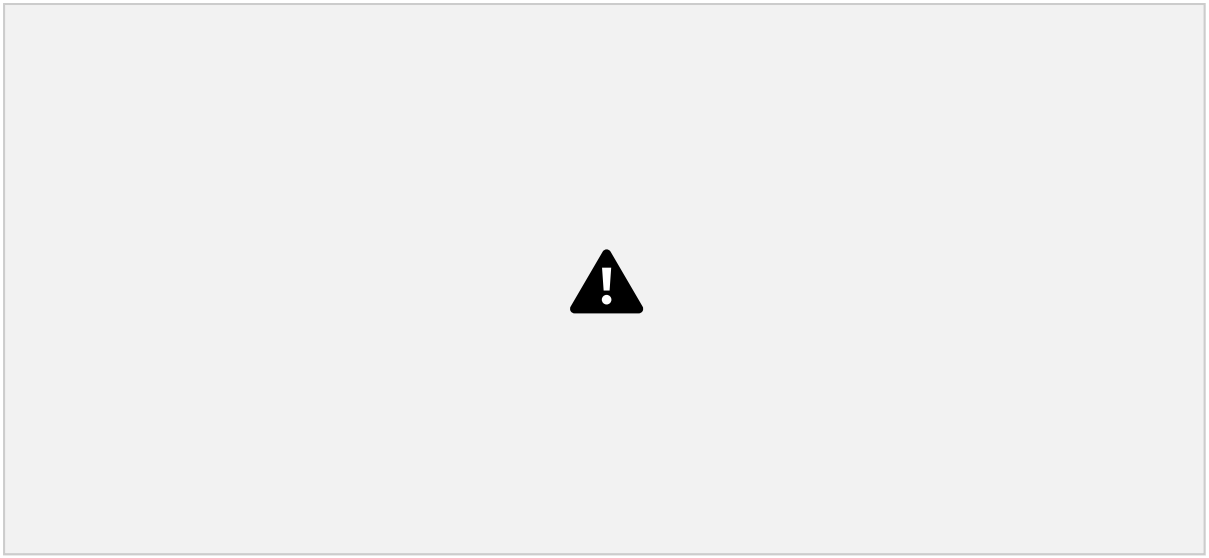
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2129877948.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

```
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
```



df.Age_Group.unique()

```
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
```

df.Hotel.unique()

```
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
```

Using the inplace=True keyword in a pandas method changes the default behaviour such that the operation on the dataframe doesn't return anything, it instead 'modifies the underlying data

df.Hotel.replace(['Ibys'],'Ibis',inplace=True)

df.FoodPreference.unique

```
<bound method Series.unique of 0 veg
1 Non-Veg
2 Veg
3 Veg
4 Vegetarian
5 Non-Veg
6 Vegetarian
7 Veg
8 Non-Veg
9 non-Veg
Name: FoodPreference, dtype: object>
```

df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)

df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)

Fillna is a Pandas function to fill the NA/NaN values with the specified method.

If column or feature is numerical continuous data then replace the missing(NaN) value by taking mean value.

If column or feature is numerical discrete data then replace the missing(NaN) value by taking median value.
If column or feature is non-numerical i.e Categorical data then replace the missing(NaN) value by taking mode value.

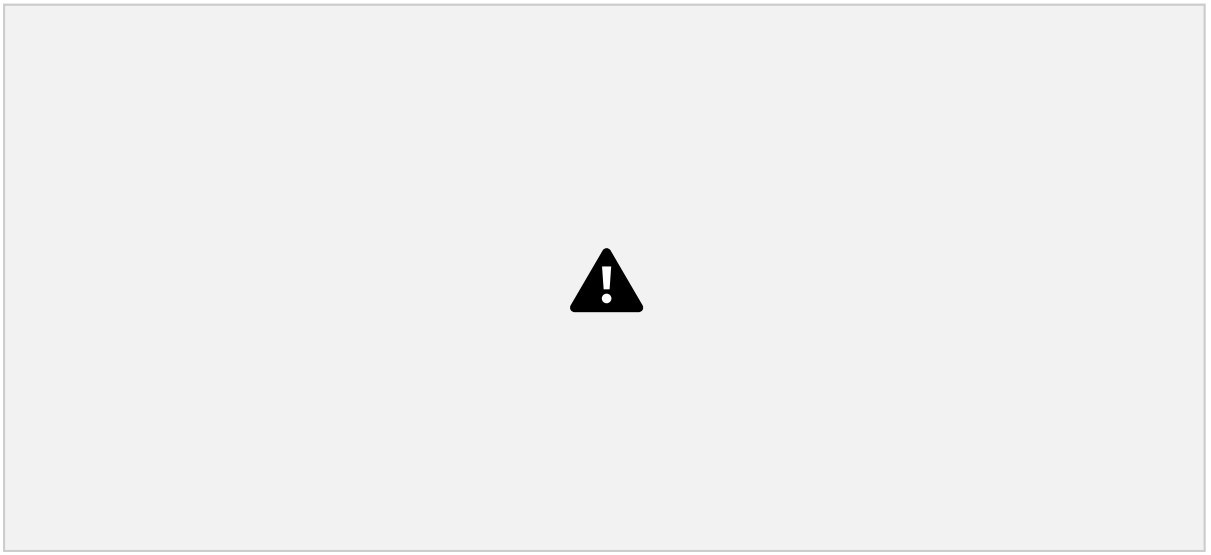
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)

df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)

df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)

df.Bill.fillna(round(df.Bill.mean()),inplace=True)

df



Exp No:**14** Experiment to understand feature scaling. Description: Understand the importance of feature scaling


```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/pre-process_datasample.csv')
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Next steps: `df.head()` Generate code with `df` toggle_off View recommended plots New interactive sheet

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40	0	
	NaN			Yes

recommended plots New interactive sheet

Next steps:
Generate code with `df` toggle_off View

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
features=df.iloc[:,:-1].values
```

```
<ipython-input-5-20665a0bbaa1>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame o
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate ob

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inpla


df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
label=df.iloc[:,-1].values
```

```
Start coding or generate with AI.
```



```
from sklearn.impute import SimpleImputer
```

```
age=SimpleImputer(strategy="mean",missing_values=np.nan)
```

```
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
```

```
age.fit(features[:,[1]])
```

```
▾ SimpleImputer i ?
SimpleImputer()
```

```
Salary.fit(features[:,[2]])
```

```
▾ SimpleImputer i ?
SimpleImputer()
```

```
SimpleImputer()
```

```
▾ SimpleImputer i ?
SimpleImputer()
```

```
features[:,[1]]=age.transform(features[:,[1]])
```

```
features[:,[2]]=Salary.transform(features[:,[2]])
```

```
features
```

```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       ['France', 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import OneHotEncoder
```

```
oh = OneHotEncoder(sparse_output=False)
```

```
Country=oh.fit_transform(features[:,[0]])
```

```
Country
```

```
array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.],
```



```
       [1., 0., 0.],
       [1., 0., 0.]])
```

```
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
```

```
final_set
```

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 48.0, 79000.0],
       [1.0, 0.0, 0.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
```

```
feat_standard_scaler
```

```
array([[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         7.58874362e-01, 7.49473254e-01],
       [-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
```

```
       -1.71150388e+00, -1.43817841e+00],
       [-1.00000000e+00, 2.00000000e+00, -6.54653671e-01,
       -1.27555478e+00, -8.91265492e-01],
       [-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
       -1.13023841e-01, -2.53200424e-01],
       [-1.00000000e+00, 2.00000000e+00, -6.54653671e-01,
       1.77608893e-01, 6.63219199e-16],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
       -5.48972942e-01, -5.26656882e-01],
       [-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
       0.00000000e+00, -1.07356980e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
       1.34013983e+00, 1.38753832e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
       1.63077256e+00, 1.75214693e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
       -2.58340208e-01, 2.93712492e-01]])
```

```
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

```
     array([[1. , 0. , 0. , 0.73913043, 0.68571429],
      [0. , 0. , 1. , 0. , 0. ],
      [0. , 1. , 0. , 0.13043478, 0.17142857],
      [0. , 0. , 1. , 0.47826087, 0.37142857],
      [0. , 1. , 0. , 0.56521739, 0.45079365],
      [1. , 0. , 0. , 0.34782609, 0.28571429],
      [0. , 0. , 1. , 0.51207729, 0.11428571],
      [1. , 0. , 0. , 0.91304348, 0.88571429],
      [1. , 0. , 0. , 1. , 1. ],
      [1. , 0. , 0. , 0.43478261, 0.54285714]])
```

```
Start coding or generate with AI.
```

Exp No:**15** Experiment to understand the data preprocessing in Data science

Description: Understand the importance of Data preprocessing in data science


```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/pre-process_datasample.csv")
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Double-click (or enter) to edit

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Country 9 non-null object
 1 Age 9 non-null float64
 2 Salary 9 non-null float64
 3 Purchased 10 non-null object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
df.Country.mode()
```

	Country
0	France

```
df.Country.mode()[0]
```

```
type(df.Country.mode())
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
~~index is not None, the resulting Series is reindexed with the index values.~~
dtype : str, numpy.dtype, or ExtensionDtype, optional
 Data type for the output Series. If not specified, this will be
 inferred from `data`.
 See the :ref:`user guide <basics.dtypes>` for more usages.
name : Hashable, default None
    The name to give to the Series
```

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
df.Age.fillna(df.Age.median(),inplace=True)
```

```
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37 0	67000 0	Yes

```
pd.get_dummies(df.Country)
```

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	True	False	False

```
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No
9	True	False	False	37 0	67000 0	Yes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Country 10 non-null object
 1 Age 10 non-null float64
 2 Salary 10 non-null float64
 3 Purchased 10 non-null object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
```



```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37	0 67000	0 1

```
Start coding or generate with AI.
```


Exp No:**15** Experiment to understand EDA-Quantitative and Qualitative analysis.

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

```
tips=sns.load_dataset('tips')
```

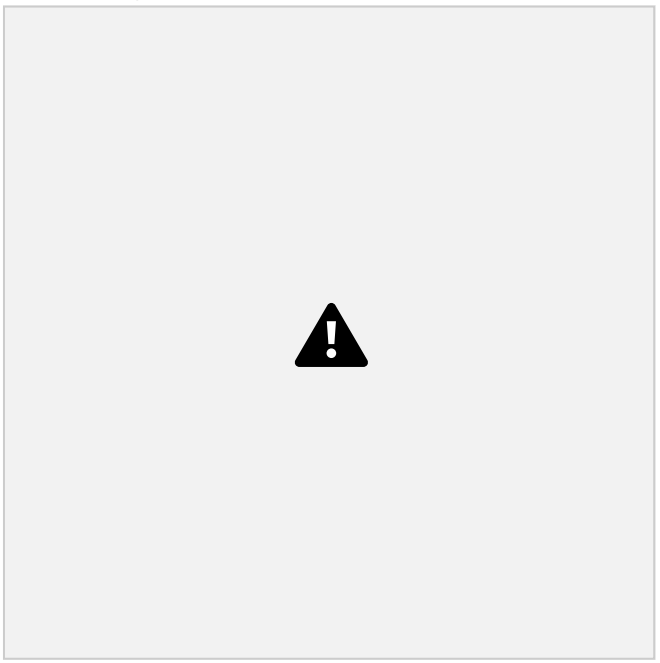
```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

add Code add Text

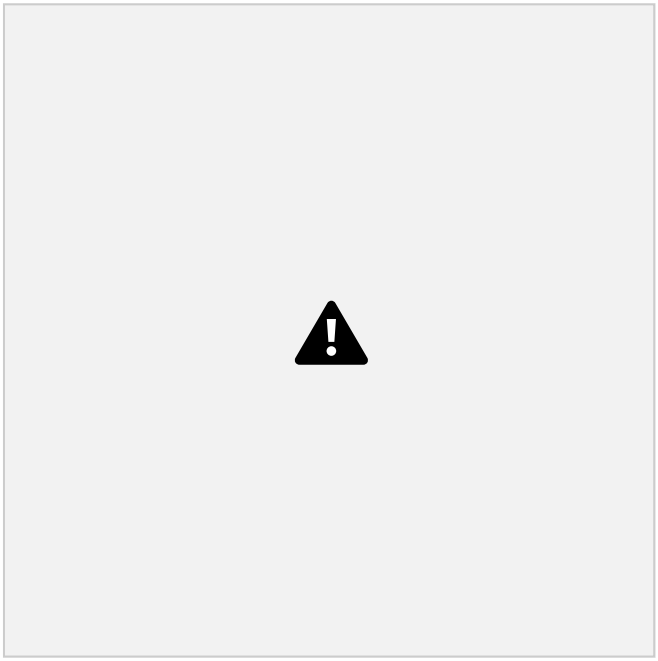
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```



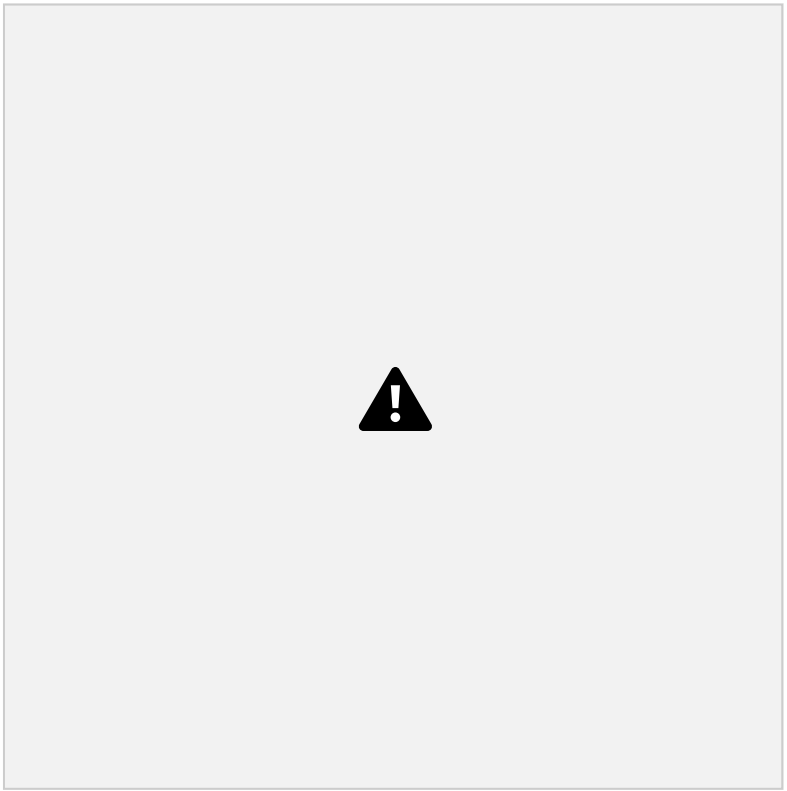
```
sns.displot(tips.total_bill,kde=False)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```



```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

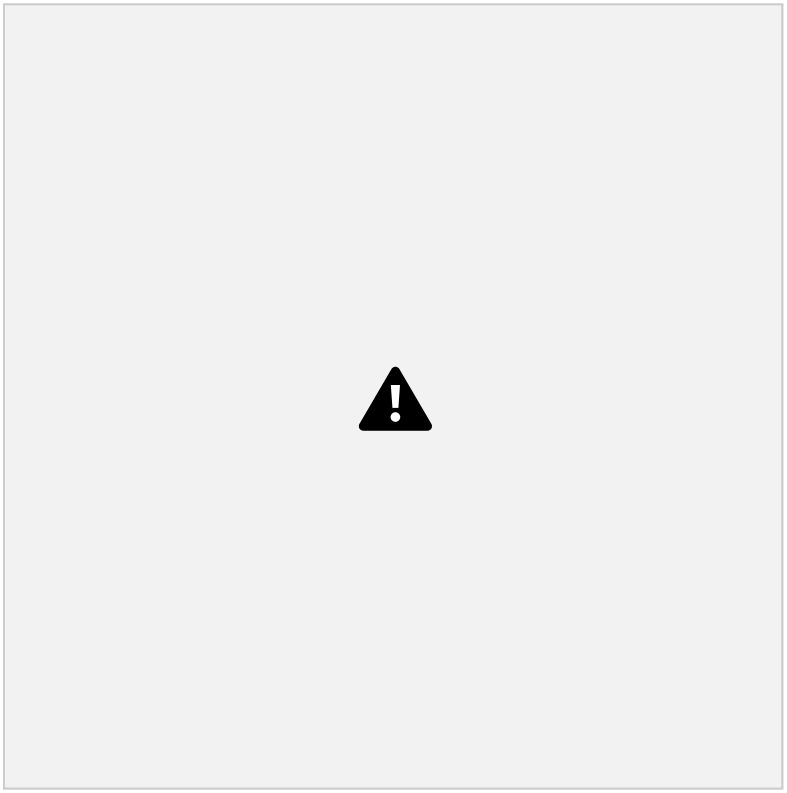
```
<seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```



```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
```

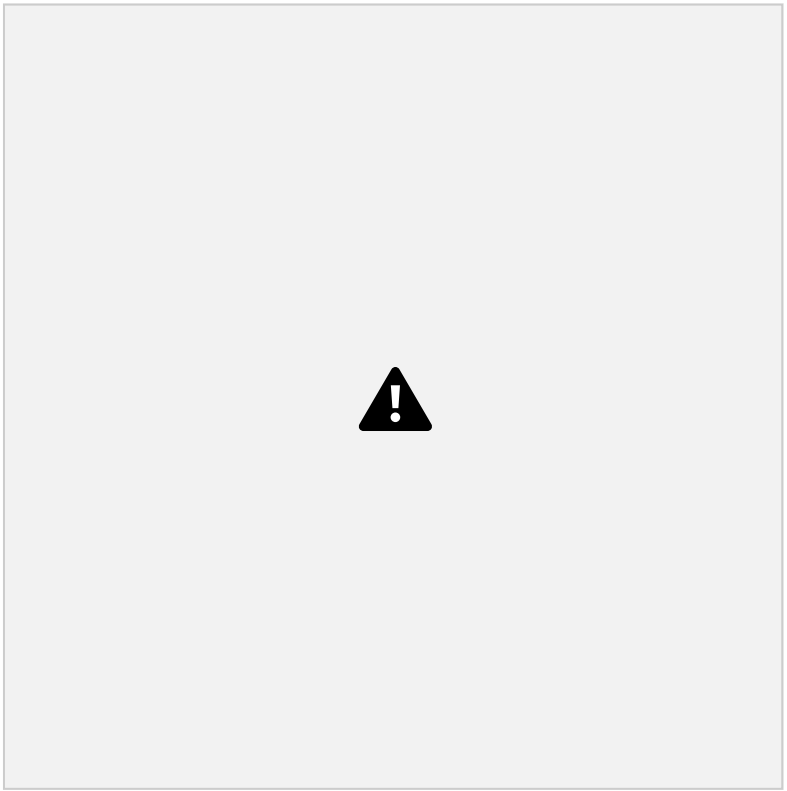


```
<seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>
```



```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
```

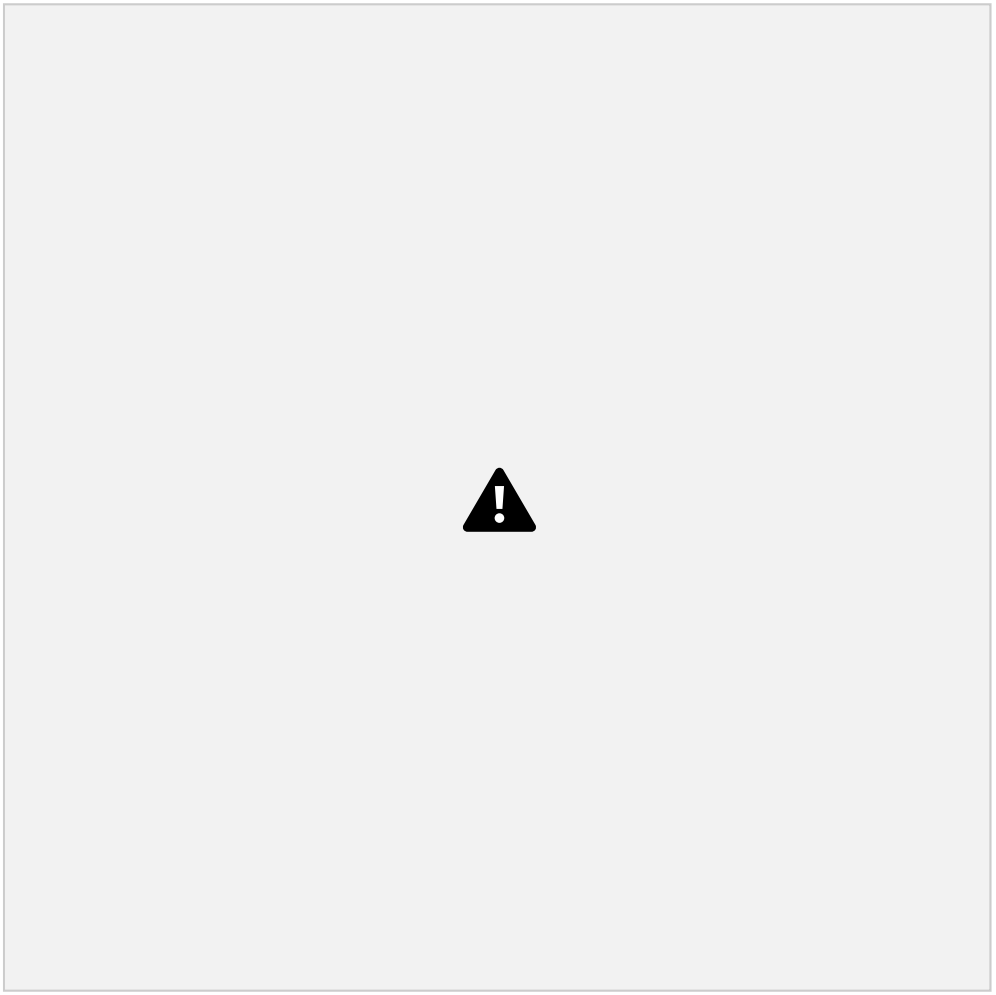
```
<seaborn.axisgrid.JointGrid at 0x79bb088f4730>
```



```
sns.pairplot(tips)
```



```
<seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```



```
tips.time.value_counts()
```

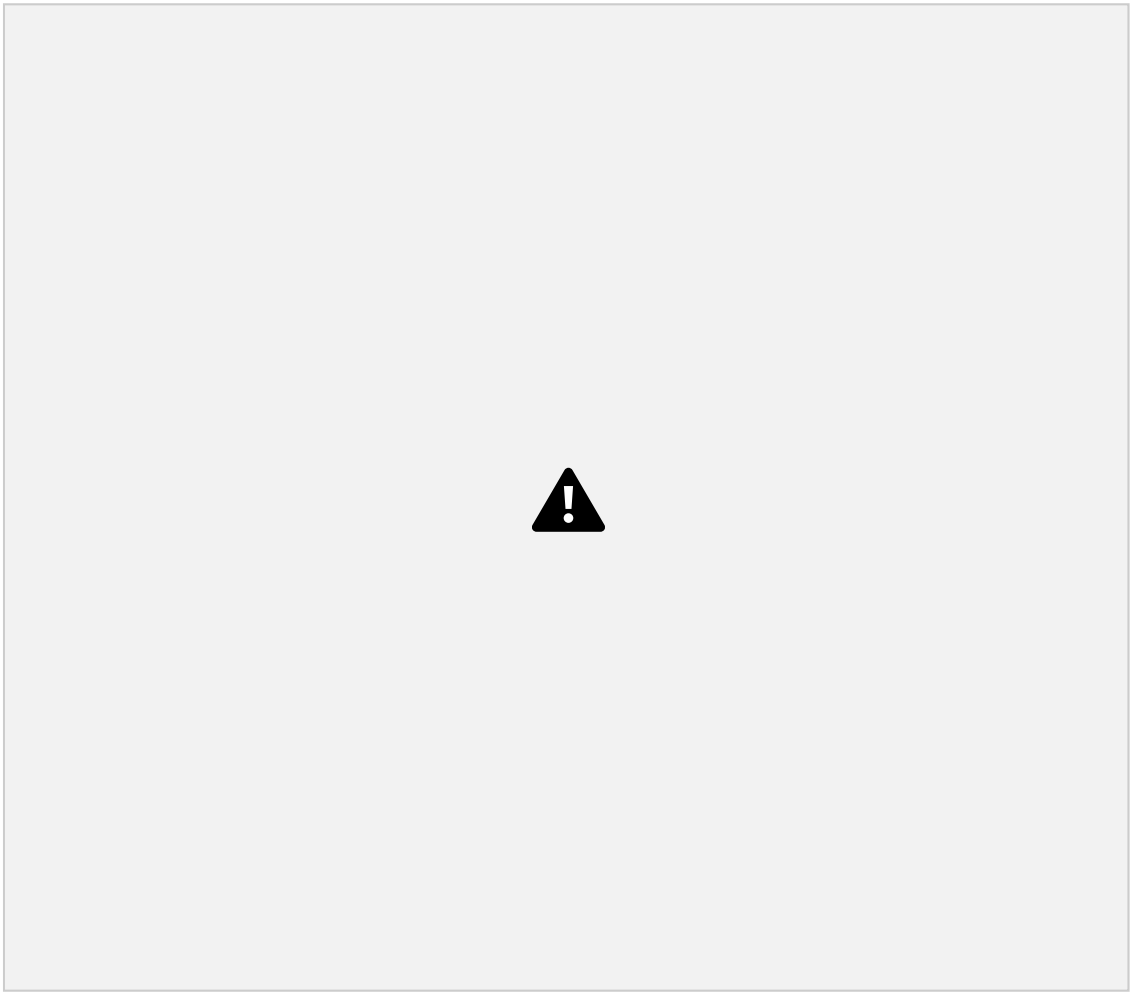
time	count
Dinner	176
Lunch	68

dtype: int64

```
sns.pairplot(tips,hue='time')
```



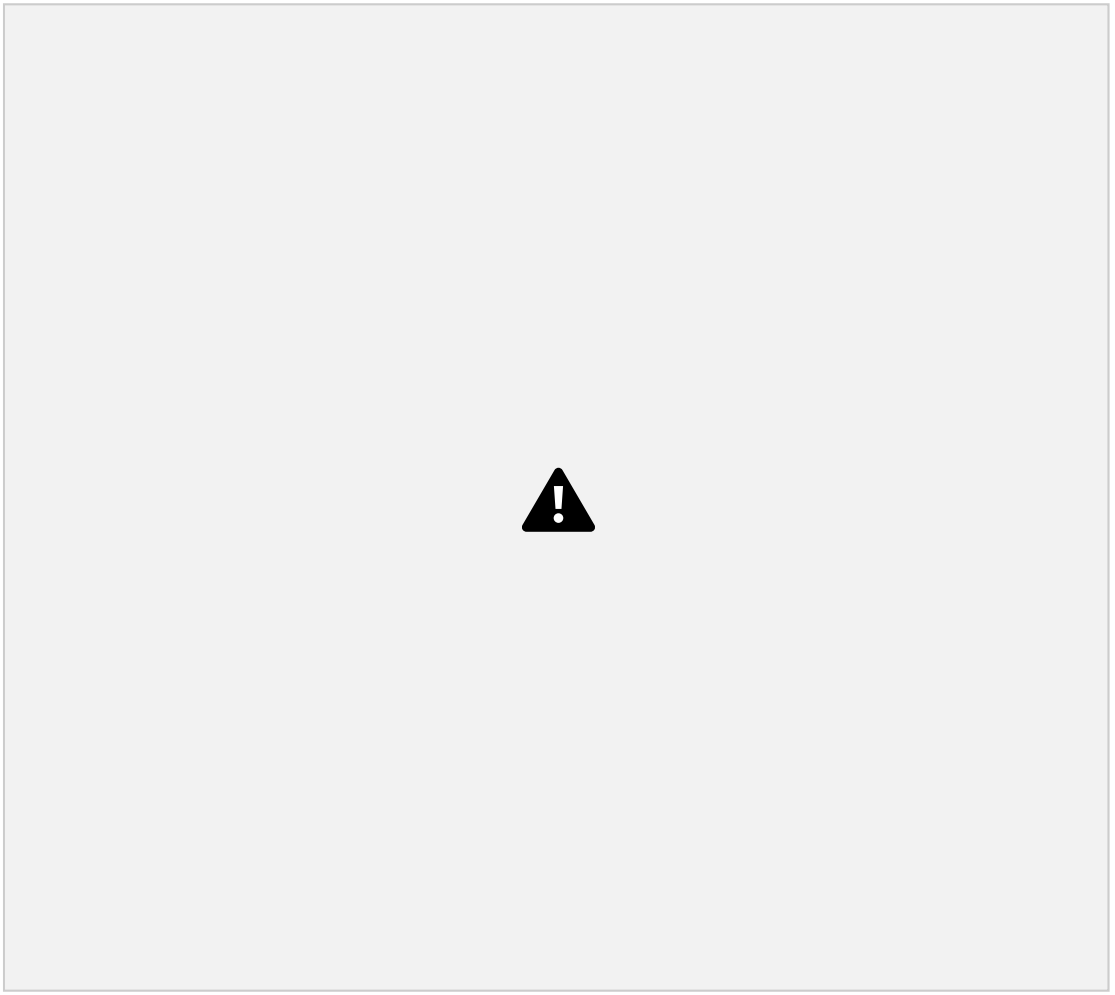
```
<seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```



```
sns.pairplot(tips,hue='day')
```

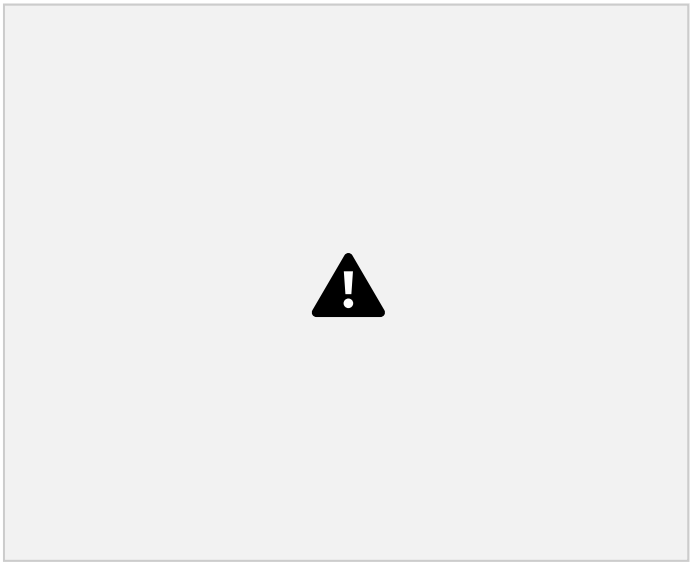


```
<seaborn.axisgrid.PairGrid at 0x79bb08f1f6a0>
```



```
sns.heatmap(tips.corr(numeric_only=True),annot=True)
```

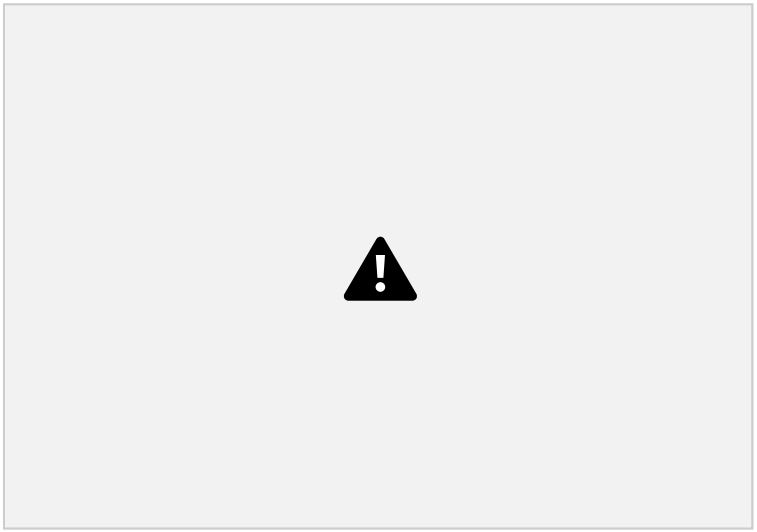
```
<Axes: >
```



```
sns.boxplot(tips.total_bill)
```

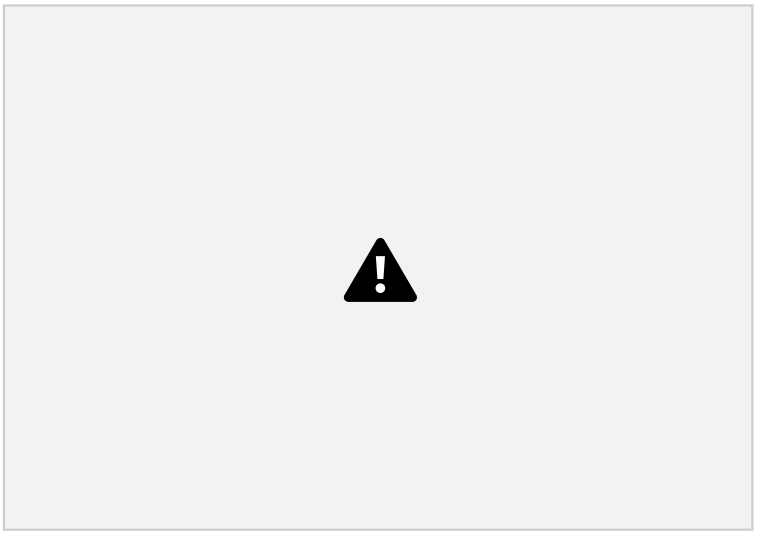


```
<Axes: ylabel='total_bill'>
```



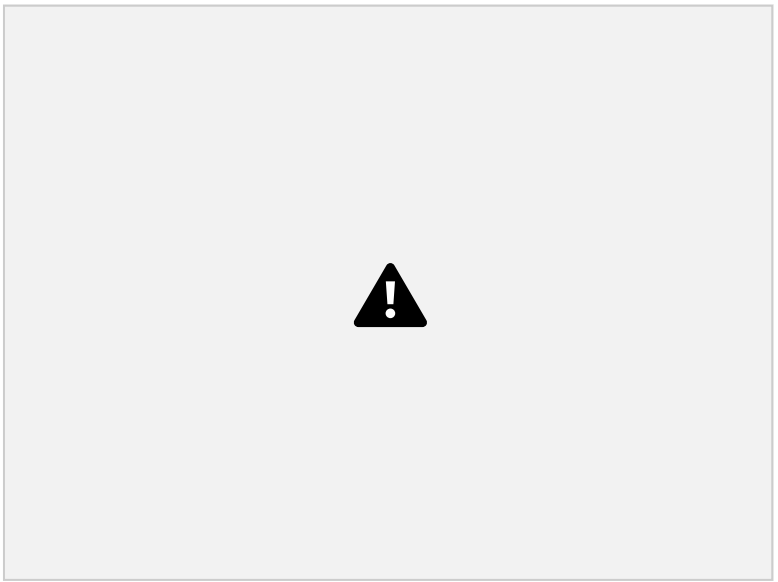
```
sns.boxplot(tips.tip)
```

```
<Axes: ylabel='tip'>
```



```
sns.countplot(tips.day)
```

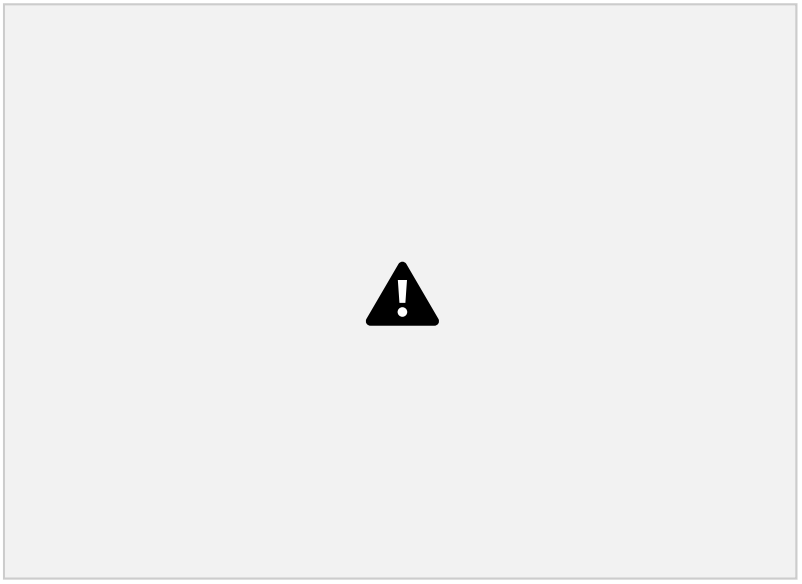
```
<Axes: xlabel='count', ylabel='day'>
```



```
sns.countplot(tips.sex)
```

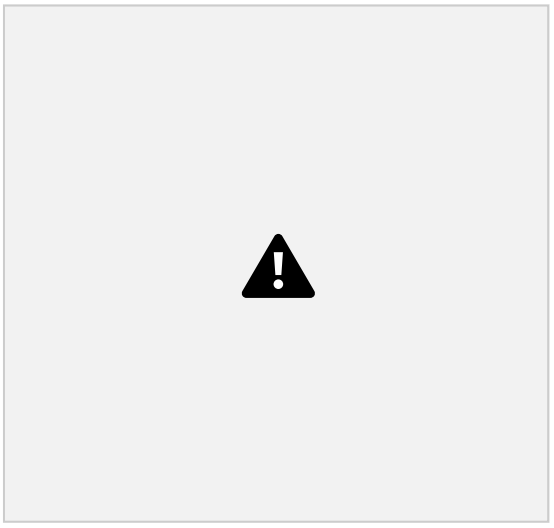


```
<Axes: xlabel='count', ylabel='sex'>
```



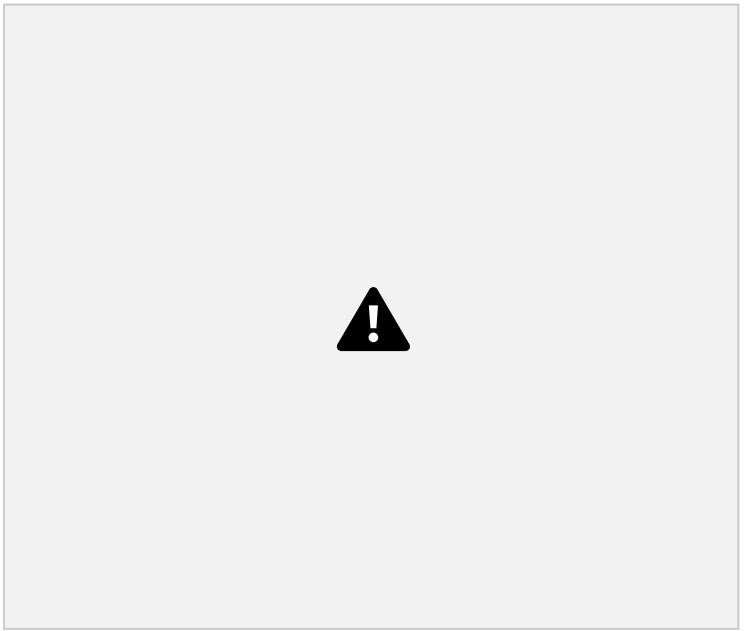
```
tips.sex.value_counts().plot(kind='pie')
```

```
<Axes: ylabel='count'>
```



```
tips.sex.value_counts().plot(kind='bar')
```

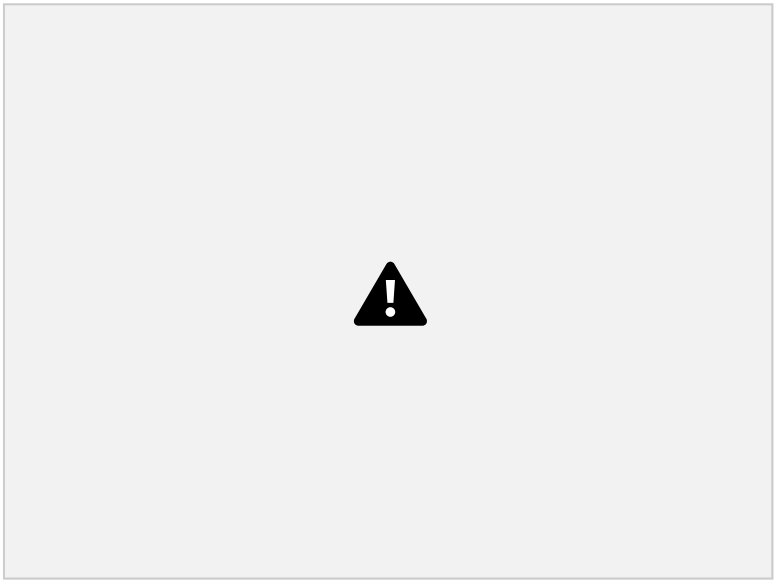
```
<Axes: xlabel='sex'>
```



```
sns.countplot(tips[tips.time=='Dinner']['day'])
```



```
<Axes: xlabel='count', ylabel='day'>
```



Exp No:**17** Experiment to understand Linear Regression for a given data set.

Description: Understand the Linear regression for the dataset given.

In [ ]: In [19]:

```
'pandas.core.frame.DataFram
e'> RangeIndex: 30 entries,
0 to 29
Data columns (total 2
columns):
# Column Non-Null Count
Dtype  --- ------
-------------- -----  0
YearsExperience 30 non-null
float64 1 Salary 30
non-null int64  dtypes:
float64(1), int64(1)
memory usage: 612.0 bytes
```

In [3]: In [4]:

```
df.dropna(inplace=True)
```

```
df.info()
```

```
<class
'pandas.core.frame.DataFram
e'> RangeIndex: 30 entries,
0 to 29
Data columns (total 2
columns):
# Column Non-Null Count
Dtype  --- ------
-------------- -----  0
YearsExperience 30 non-null
float64 1 Salary 30
non-null int64  dtypes:
float64(1), int64(1)
memory usage: 612.0 bytes
```

In [5]:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data
.csv')
df
```

```
df.info()
```

```
<class
```

```
df.describe()
```

Out[5]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [6]: In [7]: In [20]:

```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

```
from sklearn.model_selection import
train_test_split
x_train,x_test,y_train,y_test=train_test_split(
features,label,test_size=0.2,random_st
```

```
from sklearn.linear_model import
LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

Out[20]:

```
▾ LinearRegression

LinearRegression()
```

In [21]:
```
model.score(x_train,y_train)
```

Out[21]: 0.9603182547438908

In [23]:
```
model.score(x_test,y_test)
```

Out[23]: 0.9184170849214232

In [24]:
```
model.coef_
```

Out[24]: array([[9281.30847068]])

In [25]:
```
model.intercept_
```

Out[25]: array([27166.73682891])

In [26]:
```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

In [27]: In [28]:

```
Enter Years of Experience: 44
```

In [ ]: In [29]:
```
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

In [ ]:
```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))
```

Exp No:**18** Experiment to understand KNN algorithm for a given data set

Description: Understand the KNN algorithm for the dataset given.

In [1]:

In [2]:

```
df.info()
```

```
<class
'pandas.core.frame.DataFr
ame'> RangeIndex: 150
entries, 0 to 149 Data
columns (total 5
columns):
# Column Non-Null Count
Dtype  --- ------
-------------- -----  0
sepal.length 150 non-null
float64 1 sepal.width 150
non-null float64 2
petal.length 150 non-null
float64 3 petal.width 150
non-null float64 4
variety 150 non-null
object  dtypes:
float64(4), object(1)
memory usage: 6.0+ KB
```

In [3]:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Iris.csv'
)
```

```
df.variety.value_counts()
```

Out[3]:

```
Setosa 50
Versicolor 50
Virginica 50
Name: variety, dtype: int64
```

In [4]:
```
df.head(
)
```

Out[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [5]:
```
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
```

In [6]:
```
from sklearn.model_selection import
train_test_split
```

In [8]:
```
from sklearn.neighbors import
KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split
(features,label,test_size=.2,rando
model_KNN=KNeighborsClassifier(n_neighbors
=5)
model_KNN.fit(xtrain,ytrain)
```

Out[8]:
```
KNeighborsClassifier()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.